

PRICING AND HEDGING OF RENEWABLE ENERGY QUANTO OPTIONS IN CARMA MODELS

VINCENT GACHON AND SVEN KARBACH

ABSTRACT. We study quanto claims whose payoff depends on a tradable energy factor and on a second factor that captures volumetric, weather, foreign-exchange, or cross-market risk. The paper develops a coupled continuous-time autoregressive moving average (CARMA) framework in which the log-price factor and the secondary factor are driven by Lévy noises linked through a directed coupling channel. This specification preserves the interpretable low-dimensional structure of univariate CARMA models while retaining a tractable joint characteristic function and an explicit state-space representation. We show that the model embeds naturally into a multivariate CARMA formulation, which clarifies when the coupled specification is sufficient and when a full multivariate treatment is preferable.

On the pricing side we formulate the problem on a filtered space with domestic numeraire and an exogenously chosen pricing measure, which is essential because the weather or quantity factor is typically non-tradable. We derive conditional transform formulas, forward-price representations, and a Fourier valuation result for a broad class of European quanto payoffs. For the practically important case of weak Brownian coupling, we obtain a first-order quanto adjustment that isolates the contribution of the dependence channel to option values. On the hedging side we emphasize market incompleteness and characterize the variance-optimal hedge through the Galtchouk–Kunita–Watanabe (GKW) decomposition. This yields a transparent separation between the hedgeable energy price exposure and the residual hybrid risk that cannot be eliminated without additional tradable instruments.

1. INTRODUCTION

Quanto-type contracts in energy and climate-sensitive markets are hybrid claims: their value depends simultaneously on a tradable price signal and on a second factor that captures volume, temperature, irradiance, wind availability, congestion, or settlement-currency effects. In retail gas, power, and renewable generation portfolios this second factor is often the economically dominant source of non-standard exposure because demand and production volumes react to weather conditions while revenues remain tied to commodity prices. Energy quanto contracts are therefore designed to hedge joint price and quantity risk rather than either component in isolation; see [?] for the classical futures-based approach and [?] for a recent coupled electricity–temperature model.

The modeling challenge is structural. The secondary factor is often not tradable, so no-arbitrage arguments alone do not determine a unique pricing measure. At the same time, the dependence between the price factor and the secondary factor matters materially for both valuation and hedging. Simple bivariate Ornstein–Uhlenbeck specifications are appealing because they lead to closed formulas, but they impose overly rigid autocovariance structures and do not exploit the richer memory patterns observed in power prices, renewable generation, temperature, and other environmental series. In electricity markets, Lévy-driven

Key words and phrases. quanto option, CARMA process, multivariate CARMA, incomplete market, variance-optimal hedging, energy derivatives.

CARMA (continuous-time autoregressive moving average) dynamics are now a standard way of capturing mean reversion, serial dependence, and spikes while maintaining tractable state-space representations; see [?, ?, and ?]. A $\text{CARMA}(p, q)$ process is the continuous-time analogue of an $\text{ARMA}(p, q)$ time series: the autoregressive polynomial of degree p governs the mean-reversion speed and spectral shape, while the moving-average polynomial of degree $q < p$ shapes the impulse response. Driving the state equation with a Lévy process allows heavy tails, jumps, and skewness without sacrificing the linear state-space structure. The same state-space machinery is attractive in hybrid derivative problems because it supports transform methods, filtering, and dynamic hedging computations.

The present paper develops a pricing and hedging framework for energy quanto options driven by coupled CARMA factors. The word “quanto” is used here in the broad sense common in energy markets: the payoff depends on two underlying indices, one typically tradable and one often non-tradable. The classical foreign-exchange quanto is recovered when the secondary factor is itself a tradable exchange rate. We formulate the problem in a way that covers both interpretations. The starting point is a coupled state-space specification in which the secondary factor drives its own CARMA dynamics and enters the price factor through a directional coupling term. This construction preserves parsimony and economic interpretation: the secondary factor can represent a causal weather, load, or FX channel affecting the price process without forcing the entire system into a fully symmetric multivariate model.

The contribution of the paper is fivefold.

- (i) We formulate a coupled Lévy-driven CARMA model for the joint dynamics of a tradable energy factor and a secondary risk factor, and we derive its state-space, cross-covariance, and characteristic-function representations.
- (ii) We show that the coupled specification is a structured special case of a multivariate CARMA model in the sense of [?]; this clarifies when the lower-dimensional coupled form is appropriate and when a full MCARMA formulation is preferable. In particular, the multivariate viewpoint becomes compelling once several tradeable commodities, locations, or currencies are modeled jointly, or when feedback effects are empirically relevant.
- (iii) We develop a valuation framework under a domestic numeraire and a chosen pricing measure. For a broad class of European quanto payoffs we derive a Fourier representation in terms of the conditional joint characteristic function. We also obtain a first-order quanto adjustment for weak coupling, which isolates the marginal pricing impact of the dependence channel.
- (iv) We analyze hedging in the incomplete market generated by a non-tradable secondary factor. The appropriate benchmark is not perfect replication but variance-optimal hedging. We characterize the hedge by the Galtchouk–Kunita–Watanabe decomposition and show how the residual risk measures the irreducible hybrid exposure.
- (v) We calibrate the coupled CARMA framework to hourly German day-ahead electricity prices and temperature observations (2020–2025), fit Normal Inverse Gaussian driving noise, estimate the coupling parameter from cross-covariance data, validate the Fourier pricing formula against Monte Carlo simulation, and compare the resulting variance-optimal hedge against an OU benchmark in an out-of-sample 2025 backtest.

Relative to the literature on energy quantos, the novelty is therefore not a single new numerical scheme in isolation, but a coherent mathematical framework that

links CARMA dynamics, measure selection, transform pricing, and incomplete-market hedging in a single model class, together with an empirical demonstration on real hourly data.

The remainder of the paper is organized as follows. ?? recalls the state-space representation of Lévy-driven CARMA factors. ?? introduces the coupled model, derives its joint transform, and relates it to multivariate CARMA. ?? develops the pricing framework, including Fourier valuation and a weak-coupling expansion. Hedging is discussed in ?. ?? presents the empirical calibration to German electricity and temperature data. ?? reports the pricing and hedging numerical experiments. ?? highlights limitations and extensions, and ?? concludes.

2. CARMA PRELIMINARIES

We work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{M})$ satisfying the usual conditions, where $\mathbb{M}$ stands for either the historical measure $\mathbb{P}$ or a pricing measure $\mathbb{Q}$. All processes below are assumed càdlàg and adapted.

Definition 2.1. *Let $L = (L_t)_{t \geq 0}$ be an m -dimensional Lévy process under $\mathbb{M}$, let $A \in \mathbb{R}^{p \times p}$, $B \in \mathbb{R}^{p \times m}$, and let $c \in \mathbb{R}^p$. We call the output process*

$$Y_t = c^\top Z_t, \quad t \geq 0,$$

a Lévy-driven CARMA factor in state-space form if the state process $(Z_t)_{t \geq 0}$ solves

$$dZ_t = AZ_t dt + B dL_t. \quad (1)$$

When A is chosen as the companion matrix associated with a stable autoregressive polynomial of degree p , and B, c are determined by the moving-average polynomial of degree $q < p$, (1) is the standard state-space representation of a CARMA(p, q) process; see ?]. Since the driving noise L is m -dimensional while the output Y_t is scalar, this is a multi-input single-output (MISO) state-space system; the standard CARMA(p, q) process is recovered as the special case $m = 1$ with companion matrix A . In the present paper the state-space form is the primary object because it is the natural representation for pricing and hedging. The scalar output $Y_t = c^\top Z_t$ has the rational transfer function $G(z) = c^\top (zI - A)^{-1} B$, whose numerator and denominator polynomials correspond respectively to the moving-average and autoregressive components.

Assumption 2.2. *All eigenvalues of A have strictly negative real part.*

Proposition 2.3 (Variation of constants and stationarity). *Under ??, the state equation (1) has the variation-of-constants representation*

$$Z_t = e^{A(t-s)} Z_s + \int_s^t e^{A(t-u)} B dL_u, \quad 0 \leq s \leq t. \quad (2)$$

If $\mathbb{E}^{\mathbb{M}}[\log(1 + \|L_1\|)] < \infty$, then there exists a unique strictly stationary solution given by

$$Z_t = \int_{-\infty}^t e^{A(t-u)} B dL_u, \quad t \geq 0. \quad (3)$$

Proof. Equation (1) is the standard linear SDE solution. Under ??, $\|e^{At}\|$ decays exponentially. The condition $\mathbb{E}^{\mathbb{M}}[\log(1 + \|L_1\|)] < \infty$ is the minimal integrability requirement for the improper stochastic integral in (2) to converge almost surely and in L^1 ; see ?]. Under these conditions the integral gives the unique strictly stationary solution of (1). $\square$

For transform pricing we need the conditional characteristic function of the state-space output.

Proposition 2.4 (Conditional characteristic function). *Let $\psi_{\mathbb{M}} : \mathbb{R}^m \rightarrow \mathbb{C}$ denote the characteristic exponent of L under $\mathbb{M}$ (also called the Lévy symbol or cumulant generating function on the imaginary line), so that*

$$\mathbb{E}^{\mathbb{M}}[e^{i\langle u, L_t \rangle}] = e^{t\psi_{\mathbb{M}}(u)}, \quad u \in \mathbb{R}^m;$$

see [?] for background on Lévy processes and their characteristic exponents. Then for $0 \leq t \leq T$ and $u \in \mathbb{R}$,

$$\mathbb{E}^{\mathbb{M}}[e^{iuY_T} | \mathcal{F}_t] = \exp\left(iu c^\top e^{A(T-t)} Z_t + \int_t^T \psi_{\mathbb{M}}\left(u B^\top e^{A^\top(T-s)} c\right) ds\right). \quad (4)$$

Proof. By ([?]),

$$Y_T = c^\top e^{A(T-t)} Z_t + \int_t^T c^\top e^{A(T-s)} B dL_s.$$

The stochastic integral over $[t, T]$ is independent of $\mathcal{F}_t$, and its characteristic function is determined by the Lévy exponent. This yields ([?]). $\square$

Remark 2.5 (Two-sided Lévy extension). *The improper stochastic integral in ([?]) requires L to be defined on the whole line $\mathbb{R}$. A canonical construction is to take an independent copy $L^{(2)}$ of L and set*

$$L_t = \begin{cases} L_t^{(1)}, & t \geq 0, \\ -L_{(-t)-}^{(2)}, & t < 0, \end{cases}$$

where $L^{(1)}$ is the original one-sided process and $L_{t-}^{(2)} = \lim_{s \nearrow t} L_s^{(2)}$ denotes the left limit. Under this convention the integral in ([?]) is a well-defined L^1 -limit whenever $\mathbb{E}[\log(1 + \|L_1\|)] < \infty$ and A is stable; see [?] for details.

Remark 2.6. *The state-space form is not just a notational convenience. It makes explicit the transfer kernel $s \mapsto c^\top e^{As} B$, which is the object needed for transform methods, Monte Carlo simulation, and the derivation of covariance equations. It also clarifies why CARMA models remain analytically tractable under Lévy forcing: linearity reduces pricing questions to matrix exponentials and Lévy cumulants.*

3. COUPLED CARMA MODELS FOR QUANTO RISK

Convention. From this section onward, all processes, expectations, and characteristic functions are understood to be under the pricing measure $\mathbb{Q}$ unless explicitly indicated otherwise. The superscript $\mathbb{M}$ is retained only in statements that hold simultaneously under $\mathbb{P}$ and $\mathbb{Q}$, as in the second-order structure of [?]. State vectors such as $Z_t^{X, \mathbb{Q}}$ carry the measure superscript in the model equations where the distinction matters; in the pricing and hedging sections the superscript is dropped for readability since $\mathbb{Q}$ is unambiguous.

3.1. Model specification. We distinguish a *tradable* energy factor from a *secondary* factor. The secondary factor may be temperature, wind, solar irradiance, demand, an FX rate, or any other quantity entering the payoff. Denote by X_t the deseasonalized log-price factor and by Y_t the deseasonalized secondary factor. The observed spot price is $S_t = e^{\Lambda_S(t) + X_t}$, where Λ_S is a deterministic seasonal component, and the observed secondary index is $\Lambda_Y(t) + Y_t$, with deterministic component Λ_Y .

The dynamics under a generic measure $\mathbb{M} \in \{\mathbb{P}, \mathbb{Q}\}$ are

$$\begin{aligned} dZ_t^{Y, \mathbb{M}} &= A_Y^{\mathbb{M}} Z_t^{Y, \mathbb{M}} dt + B_Y dL_t^{Y, \mathbb{M}}, \\ dZ_t^{X, \mathbb{M}} &= A_X^{\mathbb{M}} Z_t^{X, \mathbb{M}} dt + B_X dL_t^{X, \mathbb{M}} + \Gamma dL_t^{Y, \mathbb{M}}, \end{aligned} \quad (5)$$

with output equations

$$X_t = c_X^\top Z_t^{X,\mathbb{M}}, \quad Y_t = c_Y^\top Z_t^{Y,\mathbb{M}}. \quad (6)$$

Here the state vectors satisfy $Z^{X,\mathbb{M}} \in \mathbb{R}^{p_X}$ and $Z^{Y,\mathbb{M}} \in \mathbb{R}^{p_Y}$, and the Lévy drivers $L^{X,\mathbb{M}} \in \mathbb{R}^{m_X}$ and $L^{Y,\mathbb{M}} \in \mathbb{R}^{m_Y}$ are independent. The *drift matrices* $A_X^{\mathbb{M}} \in \mathbb{R}^{p_X \times p_X}$ and $A_Y^{\mathbb{M}} \in \mathbb{R}^{p_Y \times p_Y}$ are measure-dependent (they absorb the effect of Girsanov or Esscher changes). The *noise-loading matrices* $B_X \in \mathbb{R}^{p_X \times m_X}$, $B_Y \in \mathbb{R}^{p_Y \times m_Y}$, the *coupling matrix* $\Gamma \in \mathbb{R}^{p_X \times m_Y}$, and the *output vectors* $c_X \in \mathbb{R}^{p_X}$, $c_Y \in \mathbb{R}^{p_Y}$ are structural parameters that do not depend on the choice of measure. This is important: the observation equation defines the *same* physical processes X_t and Y_t under every equivalent measure.

Assumption 3.1. *Throughout the paper we impose the following conditions.*

- (a) *The drift matrices $A_X^{\mathbb{M}}$ and $A_Y^{\mathbb{M}}$ are stable (all eigenvalues have strictly negative real part) under every measure $\mathbb{M}$ considered.*
- (b) *The Lévy drivers $L^{X,\mathbb{M}}$ and $L^{Y,\mathbb{M}}$ are independent and admit exponential moments: there exist $\theta, \rho > 0$ such that*

$$\mathbb{E}^{\mathbb{M}} \left[e^{\theta \|L_1^{X,\mathbb{M}}\| + \rho \|L_1^{Y,\mathbb{M}}\|} \right] < \infty,$$

which guarantees that the cumulant exponents are finite on a non-empty open strip $\mathcal{S} \subset \mathbb{R}^{m_X} \times \mathbb{R}^{m_Y}$ around the origin, enabling forward-price computations and Fourier inversion.

- (c) *There exists an equivalent probability measure $\mathbb{Q}$ such that:*
 - (i) *under $\mathbb{Q}$, the factor dynamics retain the coupled state-space form (??) with the same noise-loading matrices B_X, B_Y, Γ and output vectors c_X, c_Y , but possibly different drift matrices $A_X^{\mathbb{Q}}, A_Y^{\mathbb{Q}}$ and Lévy triplets;*
 - (ii) *the drivers $L^{X,\mathbb{Q}}$ and $L^{Y,\mathbb{Q}}$ remain independent under $\mathbb{Q}$;*
 - (iii) *discounted tradable forward prices formed from the energy factor are $\mathbb{Q}$ -martingales.*

A standard mechanism satisfying (i)–(ii) is the Esscher transform applied to the joint Lévy driver (L^X, L^Y) (see ?? below). Condition (iii) imposes no-arbitrage consistency and determines the drift adjustment of the energy factor under $\mathbb{Q}$.

Remark 3.2. *The Esscher transform preserves the coupled structure. If $L = (L^X, L^Y)$ has independent components and one applies an exponential tilt with parameter vector $(\vartheta_X, \vartheta_Y) \in \mathcal{S}$, the resulting measure $\mathbb{Q}^\vartheta$ modifies the Lévy triplet of each component independently. In particular, $L^{X,\mathbb{Q}^\vartheta}$ and $L^{Y,\mathbb{Q}^\vartheta}$ remain independent, and the linear state-space dynamics (??) are preserved with the same structural matrices B_X, B_Y, Γ . The drift matrices A_X, A_Y change only if the Esscher tilt interacts with the compensator of a jump part (which adds a drift correction); in the Gaussian case, the Esscher tilt reduces to the classical Girsanov drift shift. See ?] for details in the energy CARMA context.*

Drift adjustment and martingale calibration. Under the Esscher measure $\mathbb{Q}^\vartheta$ the cumulant exponent of L^X shifts to

$$\kappa_X^{\mathbb{Q}}(u) = \kappa_X^{\mathbb{P}}(u + \vartheta_X) - \kappa_X^{\mathbb{P}}(\vartheta_X), \quad u \in \mathcal{S}_X - \vartheta_X.$$

The tilt modifies the mean of $L^{X,\mathbb{Q}}$, which in the state equation (??) produces a drift correction that enters the last row of the companion matrix: in the CARMA(p_X, q_X) parametrisation one has

$$A_X^{\mathbb{Q}} = A_X^{\mathbb{P}} + B_X(\kappa_X^{\mathbb{P}}'(\vartheta_X) - \kappa_X^{\mathbb{P}}'(0))c_X^\top,$$

where $\kappa_X^{\mathbb{P}}'$ denotes the gradient of the cumulant exponent.

The no-arbitrage condition (iii) requires the discounted spot $e^{-r^d t} S_t$ to be a $\mathbb{Q}$ -local martingale, where $S_t = e^{\Lambda_S(t) + X_t}$. Using the forward price representation (??) this yields one scalar equation in ϑ_X :

$$\kappa_X^{\mathbb{Q}}(c_X^{\top}(-A_X^{\mathbb{Q}})^{-1}B_X) + \kappa_Y^{\mathbb{Q}}(c_X^{\top}(-A_X^{\mathbb{Q}})^{-1}\Gamma) = r^d - \dot{\Lambda}_S, \quad (7)$$

where r^d is the (constant) domestic short rate from ?? and $\dot{\Lambda}_S = d\Lambda_S/dt$ is the seasonal trend drift. For NIG drivers both $\kappa_X^{\mathbb{Q}}$ and $\kappa_Y^{\mathbb{Q}}$ are analytic and strictly monotone in their respective tilt parameters, so (??) has a unique solution ϑ_X whenever the right-hand side lies in the interior of the range of the left-hand side. The tilt parameter ϑ_Y for the secondary (non-tradable) factor remains a free parameter that encodes the market price of weather or quantity risk; in the absence of calibration instruments for that factor one may set $\vartheta_Y = 0$, which corresponds to pricing under the historical law of the secondary factor.

Remark 3.3. The directional structure in (??) is deliberate. It is appropriate when one wants the weather or quantity factor to affect the price factor but not conversely at the same horizon. This is natural for many renewable applications: temperature, wind, or irradiation shocks move expected production or demand and thereby affect power prices. The model remains parsimonious because dependence is introduced through the single matrix Γ instead of a fully unrestricted multivariate dynamics.

Remark 3.4. The classical foreign-exchange quanto fits into the same notation. If the secondary factor is the log FX rate and is itself tradable, then the pair (X, Y) describes joint commodity and currency risk. The main difference is conceptual rather than mathematical: in the tradable FX case the pricing measure is tied to domestic and foreign numeraires, whereas in the weather case the secondary factor is typically non-tradable and the market is incomplete.

3.2. Second-order structure. When the Lévy drivers are square-integrable, the coupled model admits an explicit stationary cross-covariance equation. This is useful both for calibration and for understanding the sign and decay of the quanto effect.

Proposition 3.5 (Stationary cross-covariance). *Assume that under $\mathbb{M}$ the processes $L^{X, \mathbb{M}}$ and $L^{Y, \mathbb{M}}$ are independent, square-integrable, and centered, with*

$$\Sigma_Y^{\mathbb{M}} = \text{Var}^{\mathbb{M}}(L_1^{Y, \mathbb{M}}).$$

Let $(Z_t^{X, \mathbb{M}}, Z_t^{Y, \mathbb{M}})$ denote the stationary solution of (??). Then the matrix

$$M^{\mathbb{M}} = \text{Cov}^{\mathbb{M}}(Z_t^{X, \mathbb{M}}, Z_t^{Y, \mathbb{M}})$$

is time-independent and solves the Sylvester equation

$$A_X^{\mathbb{M}} M^{\mathbb{M}} + M^{\mathbb{M}} (A_Y^{\mathbb{M}})^{\top} + \Gamma \Sigma_Y^{\mathbb{M}} B_Y^{\top} = 0. \quad (8)$$

For $h \geq 0$,

$$\text{Cov}^{\mathbb{M}}(X_{t+h}, Y_t) = c_X^{\top} e^{A_X^{\mathbb{M}} h} M^{\mathbb{M}} c_Y. \quad (9)$$

Proof. Let $N_t := Z_t^{X, \mathbb{M}} (Z_t^{Y, \mathbb{M}})^{\top}$. By Itô's formula for semimartingales,

$$dN_t = (dZ_t^{X, \mathbb{M}})(Z_t^{Y, \mathbb{M}})^{\top} + Z_t^{X, \mathbb{M}}(dZ_t^{Y, \mathbb{M}})^{\top} + d[Z^{X, \mathbb{M}}, Z^{Y, \mathbb{M}}]_t^c + \Delta N_t,$$

where $[\cdot, \cdot]^c$ denotes the continuous quadratic covariation. The continuous covariation between the X -state and the Y -state receives a contribution from the shared noise $L^{Y, \mathbb{M}}$ entering both equations through B_Y and Γ , and a zero contribution from $L^{X, \mathbb{M}}$ (which is independent of $L^{Y, \mathbb{M}}$). For the jump covariation: since the only shared noise between $Z^{X, \mathbb{M}}$ and $Z^{Y, \mathbb{M}}$ is $L^{Y, \mathbb{M}}$, simultaneous jumps at time s

satisfy $\Delta Z_s^{X,\mathbb{M}} = \Gamma \Delta L_s^{Y,\mathbb{M}}$ and $\Delta Z_s^{Y,\mathbb{M}} = B_Y \Delta L_s^{Y,\mathbb{M}}$. Therefore the expected rate of the jump covariation is

$$\mathbb{E}^{\mathbb{M}} \left[\frac{d}{dt} \sum_{s \leq t} \Delta Z_s^{X,\mathbb{M}} (\Delta Z_s^{Y,\mathbb{M}})^\top \right] = \Gamma \mathbb{E}^{\mathbb{M}} [\Delta L^{Y,\mathbb{M}} (\Delta L^{Y,\mathbb{M}})^\top] B_Y^\top = \Gamma \Sigma_Y^{\mathbb{M}} B_Y^\top,$$

where $\mathbb{E}^{\mathbb{M}} [\sum_{s \leq 1} \Delta L_s^Y (\Delta L_s^Y)^\top] = \int_{\mathbb{R}^{m_Y}} x x^\top \nu_Y(dx) = \Sigma_Y^{\mathbb{M}}$ by the Lévy–Itô decomposition and the square-integrability in (2.1). Substituting the state equations (2.1) and taking expectations in the stationary regime ($\mathbb{E}[dN_t/dt] = 0$) gives

$$A_X^{\mathbb{M}} M^{\mathbb{M}} + M^{\mathbb{M}} (A_Y^{\mathbb{M}})^\top + \Gamma \Sigma_Y^{\mathbb{M}} B_Y^\top = 0,$$

which is (2.1). Equivalently, the solution can be expressed as $M^{\mathbb{M}} = \int_0^\infty e^{A_X^{\mathbb{M}} \tau} \Gamma \Sigma_Y^{\mathbb{M}} B_Y^\top e^{(A_Y^{\mathbb{M}})^\top \tau} d\tau$, which is finite under (2.1). The lagged formula (2.1) follows from the Markov property and the variation-of-constants formula (2.1). $\square$

Remark 3.6 (Unique solvability of the Sylvester equation). *The equation $A_X^{\mathbb{M}} M + M (A_Y^{\mathbb{M}})^\top = C$ has a unique solution $M \in \mathbb{R}^{p_X \times p_Y}$ for every right-hand side C if and only if $A_X^{\mathbb{M}}$ and $-(A_Y^{\mathbb{M}})^\top$ share no common eigenvalue, i.e. $\lambda_i(A_X^{\mathbb{M}}) + \lambda_j(A_Y^{\mathbb{M}}) \neq 0$ for all i, j . Under (2.1) all eigenvalues of $A_X^{\mathbb{M}}$ and $A_Y^{\mathbb{M}}$ have strictly negative real parts, so $\text{Re}(\lambda_i(A_X^{\mathbb{M}}) + \lambda_j(A_Y^{\mathbb{M}})) < 0$ for every pair (i, j) ; in particular the sum is never zero. Hence (2.1) has a unique solution, given explicitly by*

$$M^{\mathbb{M}} = \int_0^\infty e^{A_X^{\mathbb{M}} \tau} \Gamma \Sigma_Y^{\mathbb{M}} B_Y^\top e^{(A_Y^{\mathbb{M}})^\top \tau} d\tau,$$

which is finite because both matrix exponentials decay to zero exponentially fast under (2.1).

Remark 3.7. *The cross-covariance is governed by a linear matrix equation. This is a genuine advantage of coupled CARMA models over ad hoc bivariate specifications: the sign and decay of dependence are encoded in the loading matrix Γ , the noise variance of the secondary factor, and the mean-reversion matrices A_X, A_Y . In particular, one can separate the contribution of persistence from the contribution of instantaneous coupling.*

Remark 3.8 (Spectral viewpoint). *The transfer functions associated with (2.1) are*

$$\begin{aligned} G_X(z) &= (c_X)^\top (zI - A_X)^{-1} B_X, \\ G_{XY}(z) &= (c_X)^\top (zI - A_X)^{-1} \Gamma, \\ G_Y(z) &= (c_Y)^\top (zI - A_Y)^{-1} B_Y. \end{aligned}$$

Hence the coupled model retains the rational spectral structure familiar from univariate CARMA models. This is useful for order selection and for matching empirical autocovariances or periodograms in calibration.

3.3. Conditional transform and forward prices. We now derive the key pricing object: the conditional joint transform of (X_T, Y_T) , i.e. the deseasonalized factor pair.

For $0 \leq t \leq T$, define

$$\begin{aligned} \bar{X}_t(T) &:= c_X^\top e^{A_X^{\mathbb{Q}}(T-t)} Z_t^{X,\mathbb{Q}}, \\ \bar{Y}_t(T) &:= c_Y^\top e^{A_Y^{\mathbb{Q}}(T-t)} Z_t^{Y,\mathbb{Q}}, \end{aligned}$$

as well as the deterministic loading kernels (which depend only on $T - s$, not on t separately)

$$\beta_X(s; T) := B_X^\top e^{(A_X^\mathbb{Q})^\top (T-s)} c_X, \quad (10)$$

$$\gamma_X(s; T) := \Gamma^\top e^{(A_X^\mathbb{Q})^\top (T-s)} c_X, \quad (11)$$

$$\beta_Y(s; T) := B_Y^\top e^{(A_Y^\mathbb{Q})^\top (T-s)} c_Y. \quad (12)$$

Let $\psi_X^\mathbb{Q}$ and $\psi_Y^\mathbb{Q}$ denote the characteristic exponents of $L^{X,\mathbb{Q}}$ and $L^{Y,\mathbb{Q}}$.

Proposition 3.9 (Joint transform). *Under ??, for any $u, v \in \mathbb{R}$ and $0 \leq t \leq T$,*

$$\begin{aligned} \varphi_t(u, v; T) &:= \mathbb{E}^\mathbb{Q}[e^{iuX_T + ivY_T} \mid \mathcal{F}_t] \\ &= \exp \left(iu \bar{X}_t(T) + iv \bar{Y}_t(T) + \int_t^T \psi_X^\mathbb{Q}(u \beta_X(s; T)) \, ds \right. \\ &\quad \left. + \int_t^T \psi_Y^\mathbb{Q}(u \gamma_X(s; T) + v \beta_Y(s; T)) \, ds \right). \end{aligned} \quad (13)$$

Proof. By variation of constants,

$$\begin{aligned} X_T &= \bar{X}_t(T) + \int_t^T c_X^\top e^{A_X^\mathbb{Q}(T-s)} B_X \, dL_s^{X,\mathbb{Q}} + \int_t^T c_X^\top e^{A_X^\mathbb{Q}(T-s)} \Gamma \, dL_s^{Y,\mathbb{Q}}, \\ Y_T &= \bar{Y}_t(T) + \int_t^T c_Y^\top e^{A_Y^\mathbb{Q}(T-s)} B_Y \, dL_s^{Y,\mathbb{Q}}. \end{aligned}$$

The increment of $L^{X,\mathbb{Q}}$ over $[t, T]$ is independent of both $\mathcal{F}_t$ and $L^{Y,\mathbb{Q}}$, so the conditional characteristic function factorizes into two Lévy transform terms. Evaluating those terms with the corresponding characteristic exponents yields (??). $\square$

The joint transform immediately produces the forward price of the energy factor. Let $\kappa_X^\mathbb{Q}$ and $\kappa_Y^\mathbb{Q}$ denote the cumulant exponents of $L^{X,\mathbb{Q}}$ and $L^{Y,\mathbb{Q}}$, defined on the set of arguments for which exponential moments exist.

Corollary 3.10 (Forward price representation). *Let $\mathcal{S}_X$ and $\mathcal{S}_Y$ denote the strips of regularity of the cumulant exponents $\kappa_X^\mathbb{Q}$ and $\kappa_Y^\mathbb{Q}$, i.e. the sets of real arguments on which the exponential moments are finite. Assume that for all $s \in [t, T]$, $\beta_X(s; T) \in \text{int}(\mathcal{S}_X)$ and $\gamma_X(s; T) \in \text{int}(\mathcal{S}_Y)$. A sufficient condition is that the kernel bounds $\sup_{s \in [t, T]} \|\beta_X(s; T)\|$ and $\sup_{s \in [t, T]} \|\gamma_X(s; T)\|$ are smaller than the radii of $\mathcal{S}_X$ and $\mathcal{S}_Y$; the kernel bounds are finite by stability of $A_X^\mathbb{Q}$ and $A_Y^\mathbb{Q}$, while the strip radii are determined by the Lévy measures. Then the forward price for physical delivery at time T is*

$$\begin{aligned} F(t, T) &= \mathbb{E}^\mathbb{Q}[S_T \mid \mathcal{F}_t] \\ &= \exp \left(\Lambda_S(T) + \bar{X}_t(T) + \int_t^T \kappa_X^\mathbb{Q}(\beta_X(s; T)) \, ds + \int_t^T \kappa_Y^\mathbb{Q}(\gamma_X(s; T)) \, ds \right). \end{aligned} \quad (14)$$

Proof. Because $S_T = e^{\Lambda_S(T) + X_T}$, the forward price is $\mathbb{E}^\mathbb{Q}[S_T \mid \mathcal{F}_t] = e^{\Lambda_S(T)} \mathbb{E}^\mathbb{Q}[e^{X_T} \mid \mathcal{F}_t]$. Setting $u = -i$ and $v = 0$ in (??) requires that $\varphi_t(-i, 0; T)$ is well-defined, i.e. that the argument $-i$ lies in the domain of analyticity of $u \mapsto \varphi_t(u, 0; T)$. By (??) this reduces to requiring $\beta_X(s; T) \in \text{int}(\mathcal{S}_X)$ and $\gamma_X(s; T) \in \text{int}(\mathcal{S}_Y)$ for all $s \in [t, T]$, where we use the cumulant-generating function via $\psi(-iw) = \kappa(w)$ for real w in the strip. We verify these inclusions as follows. Under ???? the kernels satisfy

$$\|\beta_X(s; T)\| \leq \|B_X\| \|c_X\| e^{\text{Re}(\lambda_{\max}(A_X^\mathbb{Q}))(T-s)}, \quad \|\gamma_X(s; T)\| \leq \|\Gamma\| \|c_X\| e^{\text{Re}(\lambda_{\max}(A_X^\mathbb{Q}))(T-s)},$$

so both are uniformly bounded on $[t, T]$ for any finite $T - t$. Under (???) the cumulant strips $\mathcal{S}_X$ and $\mathcal{S}_Y$ each contain a ball of strictly positive radius $r_X, r_Y > 0$ around the origin (since $\int_{\|x\|>1} e^{\theta\|x\|} \nu_X(dx) < \infty$ for $\theta \in (-\theta_0, \theta_0)$ with $\theta_0 \geq \min(\theta, \rho)$ from (??)). Sufficient conditions for $\beta_X(s; T) \in \text{int}(\mathcal{S}_X)$ and $\gamma_X(s; T) \in \text{int}(\mathcal{S}_Y)$ for all s are therefore

$$\|B_X\| \|c_X\| \sup_{s \in [t, T]} e^{|\lambda_{\max}|(T-s)} < r_X \quad \text{and} \quad \|\Gamma\| \|c_X\| \sup_{s \in [t, T]} e^{|\lambda_{\max}|(T-s)} < r_Y,$$

which are exactly the conditions stated before (??). Under these conditions $\psi_X^{\mathbb{Q}}(-i\beta_X(s; T)) = \kappa_X^{\mathbb{Q}}(\beta_X(s; T))$ and $\psi_Y^{\mathbb{Q}}(-i\gamma_X(s; T)) = \kappa_Y^{\mathbb{Q}}(\gamma_X(s; T))$ hold by standard analytic continuation of the Lévy exponent to its strip of definition, yielding (??). $\square$

Remark 3.11. *In electricity markets the spot is not directly tradable, while forwards and futures are. Formula (??) is therefore not just a pricing identity; it is the bridge between the CARMA spot specification and the actual hedging instruments available to the practitioner. This is one reason why the state-space representation is preferable to a purely pathwise spot specification.*

3.4. Coupled CARMA versus multivariate CARMA. The coupled specification is closely related to multivariate CARMA (MCARMA) models in the sense of [?]. Making this relation explicit is useful for both interpretation and model design.

Proposition 3.12 (MCARMA embedding). *Define the stacked state vector $Z_t^{\mathbb{Q}} \in \mathbb{R}^{p_X + p_Y}$ by*

$$Z_t^{\mathbb{Q}} = \begin{pmatrix} Z_t^{X, \mathbb{Q}} \\ Z_t^{Y, \mathbb{Q}} \end{pmatrix}, \quad A^{\mathbb{Q}} = \begin{pmatrix} A_X^{\mathbb{Q}} & 0 \\ 0 & A_Y^{\mathbb{Q}} \end{pmatrix}, \quad B = \begin{pmatrix} B_X & \Gamma \\ 0 & B_Y \end{pmatrix}.$$

Let $L_t^{\mathbb{Q}} = ((L_t^{X, \mathbb{Q}})^{\top}, (L_t^{Y, \mathbb{Q}})^{\top})^{\top}$ and

$$C = \begin{pmatrix} c_X^{\top} & 0 \\ 0 & c_Y^{\top} \end{pmatrix}.$$

Then (??)–(??) can be written as the multivariate state-space system

$$dZ_t^{\mathbb{Q}} = A^{\mathbb{Q}} Z_t^{\mathbb{Q}} dt + B dL_t^{\mathbb{Q}}, \quad \begin{pmatrix} X_t \\ Y_t \end{pmatrix} = C Z_t^{\mathbb{Q}}.$$

Hence every coupled CARMA specification is a structured MCARMA model. Conversely, an MCARMA model with block-diagonal state matrix and block-triangular noise loading recovers the coupled model.

Proof. This is immediate from block-matrix multiplication. $\square$

Remark 3.13. *Coupled CARMA is a restricted MCARMA specification. The restriction is valuable when the data suggest a clear direction of dependence, because it reduces the parameter count and preserves interpretability. A full MCARMA formulation is preferable when several prices, locations, or currencies are jointly tradable, or when feedback effects cannot be ignored. It is also useful when one wants positivity or cone constraints on the multivariate state, as in [?]. In those situations the multivariate formulation is more elegant and often more robust, but it is also more demanding to calibrate.*

4. PRICING OF ENERGY QUANTO OPTIONS

4.1. Market setup, numeraires, and payoff class.

Assumption 4.1. *The domestic short rate $(r_t^d)_{t \geq 0}$ is deterministic and bounded, so the domestic money-market account $B_t^d = \exp(\int_0^t r_u^d du)$ is a known deterministic function of time. This assumption decouples interest-rate risk from hybrid energy-weather risk and can be relaxed by appending a stochastic rate factor to the state vector without altering the CARMA structure.*

We focus on European claims maturing at time T . We adopt the convention that the payoff is stated in terms of the *observed* levels: the spot price $S_T = e^{\Lambda_S(T) + X_T}$ and the observed secondary index $\hat{Y}_T := \Lambda_Y(T) + Y_T$. Their discounted domestic payoff is

$$\tilde{H} = (B_T^d)^{-1} g(S_T, \hat{Y}_T),$$

where $g : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$ is measurable and integrable under $\mathbb{Q}$. The practically most important case is

$$g(s, y) = (s - K_S)^+ h(y), \quad (15)$$

where h is a weather or quantity leg. Typical examples are

$$h(y) = (K_Y - y)^+, \quad h(y) = (y - K_Y)^+, \quad h(y) = \mathbb{1}_{\{y > K_Y\}}.$$

Remark 4.2. *The pricing measure $\mathbb{Q}$ is the one specified in [1]. In particular, it makes discounted energy forward prices martingales and preserves the coupled state-space form with the same structural matrices. The choice of $\mathbb{Q}$ is considered part of the model specification.*

Remark 4.3. *In hybrid energy-weather markets the pricing measure is not uniquely determined by no-arbitrage if the secondary factor is non-tradable. One may take $\mathbb{Q}$ to coincide with the historical law of the secondary factor, or introduce a risk premium through an Esscher or state-dependent tilt. The CARMA state-space structure survives under many such measure changes on the traded side; compare the discussion of risk premia in [2]. By contrast, in a classical FX-quanto with tradable foreign asset and FX rate, domestic and foreign numeraires give a sharper no-arbitrage restriction; see [3].*

Many OTC energy quantos settle over a delivery period rather than a single date. If the contract is a discrete strip payoff of the form

$$H = \sum_{k=1}^n w_k g(S_{t_k}, \hat{Y}_{t_k}),$$

linearity of expectation reduces pricing to a finite sum of one-date problems. The formulas below therefore remain the basic building block for delivery-period claims.

4.2. Fourier valuation. The coupled CARMA model is particularly attractive when pricing is performed in transform space. To separate deterministic seasonality from the stochastic factor dynamics, define the factor-space payoff

$$g_T(x, y) := g(e^{\Lambda_S(T) + x}, \Lambda_Y(T) + y),$$

and the damped payoff

$$g_{T,\alpha}(x, y) := e^{-\alpha_1 x - \alpha_2 y} g_T(x, y),$$

for a damping vector $\alpha = (\alpha_1, \alpha_2) \in \mathbb{R}^2$. Denote by

$$\widehat{g_{T,\alpha}}(u, v) = \int_{\mathbb{R}^2} e^{-i(ux + vy)} g_{T,\alpha}(x, y) dx dy$$

its Fourier transform.

Proposition 4.4 (Fourier pricing formula). *Suppose there exists $\alpha \in \mathbb{R}^2$ such that:*

- (i) $g_{T,\alpha} \in L^1(\mathbb{R}^2)$, so that $\widehat{g_{T,\alpha}}$ exists and is bounded;
- (ii) the conditional moment condition

$$\mathbb{E}^{\mathbb{Q}}[e^{\alpha_1 X_T + \alpha_2 Y_T} | g_T(X_T, Y_T) | \mathcal{F}_t] < \infty$$

holds, which ensures that the transform $\varphi_t(u - i\alpha_1, v - i\alpha_2; T)$ is well-defined;

- (iii) *the product $(u, v) \mapsto |\widehat{g_{T,\alpha}}(u, v)| \cdot |\varphi_t(u - i\alpha_1, v - i\alpha_2; T)|$ is integrable over $\mathbb{R}^2$, which justifies the Fubini interchange.*

Conditions (i)–(iii) are satisfied, for example, when the drivers are of NIG or Variance-Gamma type and the damping vector α lies in the interior of the joint strip of regularity. Then the time- t price of the European quanto claim is

$$V_t = \frac{B_t^d}{B_T^d} \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \widehat{g_{T,\alpha}}(u, v) \varphi_t(u - i\alpha_1, v - i\alpha_2; T) du dv. \quad (16)$$

Proof. Since $g_{T,\alpha} \in L^1(\mathbb{R}^2)$, Fourier inversion gives

$$g_T(x, y) = \frac{e^{\alpha_1 x + \alpha_2 y}}{(2\pi)^2} \int_{\mathbb{R}^2} e^{i(ux + vy)} \widehat{g_{T,\alpha}}(u, v) du dv.$$

Substituting $x = X_T$, $y = Y_T$, taking conditional expectations, and interchanging expectation and integration by Fubini yields (??), because

$$\mathbb{E}^{\mathbb{Q}}[e^{(\alpha_1 + iu)X_T + (\alpha_2 + iv)Y_T} | \mathcal{F}_t] = \varphi_t(u - i\alpha_1, v - i\alpha_2; T).$$

□

Remark 4.5. *The advantage of (??) is that all model dependence enters through the transform φ_t , which is explicit by ?? . Once the transform of the damped payoff is available, pricing can be implemented by numerical quadrature or by Carr–Madan type FFT methods [?]. For product payoffs such as $(S_T - K_S)^+(K_Y - \hat{Y}_T)^+$, the damped transform factorizes into a call part and a one-dimensional transform of the secondary leg.*

Remark 4.6 (Verification of inversion conditions for the standard quanto call). *We verify conditions (i)–(iii) of ?? for the payoff $g(s, y) = (s - K_S)^+ h(y)$ with $h(y) = (K_Y - y)^+$ and damping vector $\alpha = (\alpha_1, \alpha_2)$ with $\alpha_1 > 1$ and $\alpha_2 > 0$.*

Condition (i). $|g_{T,\alpha}(x, y)| \leq e^{(\alpha_1 - 1)x} e^{-\alpha_2 y} |h(\Lambda_Y(T) + y)|$. Since $h(y) = (K_Y - y)^+$ is bounded by $|K_Y| + |y|$, the product $e^{(\alpha_1 - 1)x} e^{-\alpha_2 y} (1 + |y|)$ is integrable over $\mathbb{R}^2$ for $\alpha_1 > 1$ and $\alpha_2 > 0$, so $g_{T,\alpha} \in L^1(\mathbb{R}^2)$.

Condition (ii). The moment condition $\mathbb{E}^{\mathbb{Q}}[e^{\alpha_1 X_T + \alpha_2 Y_T} | g_T | \mathcal{F}_t] < \infty$ requires $\alpha_1 \beta_X(s; T) \in \text{int}(\mathcal{S}_X)$ and $(\alpha_1 \gamma_X(s; T) + \alpha_2 \beta_Y(s; T)) \in \text{int}(\mathcal{S}_Y)$ for all $s \in [t, T]$; this follows from the strip argument of ?? applied with the scaled damping parameters, using the kernel bounds and the exponential moment condition ????

Condition (iii). For the call–put product payoff the Fourier transform of the damped payoff satisfies $|\widehat{g_{T,\alpha}}(u, v)| = O((u^2 + v^2)^{-1})$ as $\|(u, v)\| \rightarrow \infty$, which follows from the standard decay of Fourier transforms of piecewise smooth functions. For NIG drivers, the characteristic function $\varphi_t(u - i\alpha_1, v - i\alpha_2; T)$ decays superpolynomially in $\|(u, v)\|$ because the NIG cumulant exponent $\kappa_{\text{NIG}}(w) \propto -\delta \sqrt{\alpha^2 - (\beta + w)^2} + \delta \sqrt{\alpha^2 - \beta^2}$ grows as $\delta |w|$ for large $|w|$, producing exponential decay of $|\varphi_t|$ in u and v . The product $|\widehat{g_{T,\alpha}}| |\varphi_t|$ is therefore integrable over $\mathbb{R}^2$.

4.3. A weak-coupling quanto adjustment. The full Fourier formula is general but not always the most transparent object for economic interpretation. To isolate the effect of dependence, it is useful to study the first derivative of the price with respect to the coupling strength. We therefore specialize temporarily to the following diffusion-type dependence channel on the interval $[t, T]$:

$$\begin{aligned} dZ_u^Y &= A_Y Z_u^Y du + B_Y dW_u, \\ dZ_u^{X,\lambda} &= A_X Z_u^{X,\lambda} du + B_X dL_u^X + \lambda \Gamma dW_u, \quad u \in [t, T], \end{aligned} \quad (17)$$

where W is a standard Brownian motion independent of the Lévy driver L^X , and the state at time t is kept fixed when differentiating with respect to λ . In this Gaussian specialization the secondary factor is driven by a scalar Brownian motion, so $B_Y \in \mathbb{R}^{p_Y}$ is the diffusion column vector (the matrix B_Y in the Lévy-driven model restricted to a single Brownian component). We suppress superscripts $\mathbb{Q}$ for readability throughout this subsection. This framework allows the Stein-type arguments used in the proof; in the fully general Lévy case the leading-order quanto adjustment follows from a similar but more involved cumulant-expansion argument.

Lemma 4.7 (Stein identity for jointly Gaussian variables). *Let (X, Y) be jointly normally distributed with $\mathbb{E}^{\mathbb{Q}}[X] = 0$ and $\mathbb{E}^{\mathbb{Q}}[Y] = \mu_Y$. If $h : \mathbb{R} \rightarrow \mathbb{R}$ is absolutely continuous and $\mathbb{E}^{\mathbb{Q}}[|h'(Y)|] < \infty$, then*

$$\mathbb{E}^{\mathbb{Q}}[X h(Y)] = \text{Cov}^{\mathbb{Q}}(X, Y) \cdot \mathbb{E}^{\mathbb{Q}}[h'(Y)].$$

Proof. Write $X = \rho(\sigma_X/\sigma_Y)(Y - \mu_Y) + \varepsilon$ where $\rho = \text{Cov}(X, Y)/(\sigma_X \sigma_Y)$ is the correlation and $\varepsilon \perp Y$ is the regression residual (also normally distributed with mean zero). Then

$$\mathbb{E}^{\mathbb{Q}}[X h(Y)] = \frac{\text{Cov}^{\mathbb{Q}}(X, Y)}{\sigma_Y^2} \mathbb{E}^{\mathbb{Q}}[(Y - \mu_Y)h(Y)] + \mathbb{E}^{\mathbb{Q}}[\varepsilon] \mathbb{E}^{\mathbb{Q}}[h(Y)] = \frac{\text{Cov}^{\mathbb{Q}}(X, Y)}{\sigma_Y^2} \mathbb{E}^{\mathbb{Q}}[(Y - \mu_Y)h(Y)].$$

Integration by parts for the Gaussian density gives $\mathbb{E}^{\mathbb{Q}}[(Y - \mu_Y)h(Y)] = \sigma_Y^2 \mathbb{E}^{\mathbb{Q}}[h'(Y)]$, which yields the result. $\square$

Theorem 4.8 (First-order quanto adjustment). *Consider the payoff $g(s, y) = (s - K_S)^+ h(y)$, where $h : \mathbb{R} \rightarrow \mathbb{R}$ is absolutely continuous with $\mathbb{E}^{\mathbb{Q}}[|h'(\hat{Y}_T)| \mid \mathcal{F}_t] < \infty$. Under (??), define*

$$X_T^\lambda = (c_X)^\top Z_T^{X,\lambda}, \quad Y_T = (c_Y)^\top Z_T^Y, \quad S_T^\lambda = e^{\Lambda_S(T) + X_T^\lambda}, \quad \hat{Y}_T = \Lambda_Y(T) + Y_T, \quad \Xi_{t,T} := (c_X)^\top \int_t^T e^{A_X(T-u)} \Gamma dW_u.$$

Then $X_T^\lambda = X_T^0 + \lambda \Xi_{t,T}$ and the option value

$$V_t^\lambda = \frac{B_t^d}{B_T^d} \mathbb{E}^{\mathbb{Q}} \left[(S_T^\lambda - K_S)^+ h(\hat{Y}_T) \mid \mathcal{F}_t \right]$$

is differentiable at $\lambda = 0$ with

$$\left. \frac{\partial V_t^\lambda}{\partial \lambda} \right|_{\lambda=0} = \frac{B_t^d}{B_T^d} \mathbb{E}^{\mathbb{Q}} \left[S_T^0 \mathbb{1}_{\{S_T^0 > K_S\}} \mid \mathcal{F}_t \right] \text{Cov}_t^{\mathbb{Q}}(\Xi_{t,T}, Y_T) \mathbb{E}^{\mathbb{Q}}[h'(\hat{Y}_T) \mid \mathcal{F}_t]. \quad (18)$$

Proof. By linearity of (??),

$$X_T^\lambda = X_T^0 + \lambda \Xi_{t,T},$$

where $X_T^0 = \bar{X}_t(T) + \int_t^T \beta_X(s; T)^\top dL_s^X$ is independent of W , and $\Xi_{t,T} = (c_X)^\top \int_t^T e^{A_X(T-u)} \Gamma dW_u$ is a centered Gaussian random variable with finite variance. To differentiate V_t^λ with respect to λ , note that for $\lambda \neq \lambda^* := (\log K_S - \Lambda_S(T) - X_T^0)/\Xi_{t,T}$ (i.e. away from the at-the-money boundary $S_T^\lambda = K_S$),

$$\frac{\partial}{\partial \lambda} (S_T^\lambda - K_S)^+ h(\hat{Y}_T) = S_T^\lambda \mathbb{1}_{\{S_T^\lambda > K_S\}} \Xi_{t,T} h(\hat{Y}_T).$$

The set $\{\lambda = \lambda^*\}$ has Lebesgue measure zero in λ , and for fixed λ the set $\{\omega : S_T^\lambda(\omega) = K_S\}$ has $\mathbb{Q}$ -probability zero because X_T^λ has a density (it contains the Gaussian component $\lambda \Xi_{t,T}$). Hence the right-hand derivative exists $\mathbb{Q}$ -a.s. for every λ . The interchange of derivative and conditional expectation is justified by dominated convergence: for $|\lambda| \leq 1$, the integrand is bounded in absolute value by $\sup_{\lambda \in [-1,1]} S_T^\lambda \cdot |\Xi_{t,T}| \cdot |h(\hat{Y}_T)|$, and the integrability of this dominating function follows from the Gaussianity of $\Xi_{t,T}$, the exponential moment conditions in (2.10) applied to L^X , and the assumption $\mathbb{E}^\mathbb{Q}[|h'(\hat{Y}_T)| \mid \mathcal{F}_t] < \infty$. Hence

$$\left. \frac{\partial V_t^\lambda}{\partial \lambda} \right|_{\lambda=0} = \frac{B_t^d}{B_T^d} \mathbb{E}^\mathbb{Q} \left[S_T^0 \mathbb{1}_{\{S_T^0 > K_S\}} \Xi_{t,T} h(\hat{Y}_T) \mid \mathcal{F}_t \right].$$

At $\lambda = 0$, the process $S_T^0 = e^{\Lambda_S(T) + X_T^0}$ depends only on the increment of L^X over $[t, T]$, which is independent of $\mathcal{F}_t$ and of the Brownian motion W . Since $(\Xi_{t,T}, Y_T)$ is $\sigma(W_u - W_t : u \in [t, T])$ -measurable and hence independent of S_T^0 conditional on $\mathcal{F}_t$, we obtain

$$\left. \frac{\partial V_t^\lambda}{\partial \lambda} \right|_{\lambda=0} = \frac{B_t^d}{B_T^d} \mathbb{E}^\mathbb{Q} \left[S_T^0 \mathbb{1}_{\{S_T^0 > K_S\}} \mid \mathcal{F}_t \right] \mathbb{E}^\mathbb{Q} \left[\Xi_{t,T} h(\hat{Y}_T) \mid \mathcal{F}_t \right].$$

Conditionally on $\mathcal{F}_t$, the pair $(\Xi_{t,T}, \hat{Y}_T)$ is jointly Gaussian. Indeed, $\Xi_{t,T}$ is a Gaussian stochastic integral and $Y_T = \bar{Y}_t(T) + c_Y^\top \int_t^T e^{A_Y(T-s)} B_Y dW_s$ is a Gaussian functional of the same Brownian motion, so $\hat{Y}_T$ is an affine shift of a Gaussian variable. Moreover, $\mathbb{E}^\mathbb{Q}[\Xi_{t,T} \mid \mathcal{F}_t] = 0$. (2.11) (applied conditionally on $\mathcal{F}_t$ to the pair $(\Xi_{t,T}, \hat{Y}_T - \Lambda_Y(T))$, which is jointly Gaussian and centered in its first component) therefore yields

$$\mathbb{E}^\mathbb{Q} \left[\Xi_{t,T} h(\hat{Y}_T) \mid \mathcal{F}_t \right] = \text{Cov}_t^\mathbb{Q}(\Xi_{t,T}, Y_T) \mathbb{E}^\mathbb{Q} \left[h'(\hat{Y}_T) \mid \mathcal{F}_t \right],$$

which is valid whenever h is absolutely continuous and $\mathbb{E}^\mathbb{Q}[|h'(\hat{Y}_T)| \mid \mathcal{F}_t] < \infty$. Substituting into the previous line gives (2.12). $\square$

Corollary 4.9 (Heating payoff). *If $h(y) = (K_Y - y)^+$, then $h'(y) = -\mathbb{1}_{\{y < K_Y\}}$ for almost every y , and (2.12) becomes*

$$\left. \frac{\partial V_t^\lambda}{\partial \lambda} \right|_{\lambda=0} = -\frac{B_t^d}{B_T^d} \mathbb{E}^\mathbb{Q} \left[S_T^0 \mathbb{1}_{\{S_T^0 > K_S\}} \mid \mathcal{F}_t \right] \text{Cov}_t^\mathbb{Q}(\Xi_{t,T}, Y_T) \mathbb{Q}(\hat{Y}_T < K_Y \mid \mathcal{F}_t). \quad (19)$$

Remark 4.10. Formula (2.12) makes the sign of the quanto adjustment transparent. If a positive weather innovation increases the energy price factor, so that $\text{Cov}_t^\mathbb{Q}(\Xi_{t,T}, Y_T) > 0$, then a heating-style payoff $(K_Y - \hat{Y}_T)^+$ receives a negative first-order correction. This matches economic intuition: the payoff is largest when the secondary factor is low, but positive coupling shifts the energy call value toward states where that leg is less likely to be in the money.

Remark 4.11 (Deterministic covariance and closed-form kernel). *The quantity $\text{Cov}_t^\mathbb{Q}(\Xi_{t,T}, Y_T)$ in (2.12)–(2.13) is deterministic given the system matrices and the time-to-maturity $T - t$. Indeed, both $\Xi_{t,T}$ and $Y_T - \bar{Y}_t(T)$ are Gaussian stochastic integrals with deterministic kernels, so*

$$\text{Cov}_t^\mathbb{Q}(\Xi_{t,T}, Y_T) = \int_t^T c_X^\top e^{A_X(T-u)} \Gamma B_Y^\top e^{A_Y^\top(T-u)} c_Y du.$$

This integral admits a closed-form evaluation in terms of the Sylvester operator $\hat{D}_{XY}(M) := A_X M + M A_Y^\top$ introduced in (2.14). Define the function $P(\tau) := e^{A_X \tau} \Gamma B_Y^\top e^{A_Y^\top \tau}$,

so that $dP(\tau)/d\tau = \widehat{D}_{XY}(P(\tau))$. Integrating from 0 to $T - t$ and applying the operator inverse (which exists by ??) gives

$$\text{Cov}_t^{\mathbb{Q}}(\Xi_{t,T}, Y_T) = c_X^{\top} \widehat{D}_{XY}^{-1} (e^{A_X(T-t)} \Gamma B_Y^{\top} e^{A_Y^{\top}(T-t)} - \Gamma B_Y^{\top}) c_Y. \quad (20)$$

For numerical computation one evaluates the $2p_X$ -by- $2p_X$ (or $2p_Y$ -dimensional) matrix exponential of the block matrix $\begin{pmatrix} A_X & \Gamma B_Y^{\top} \\ 0 & A_Y^{\top} \end{pmatrix} (T - t)$ and reads off the off-diagonal block, which is the standard Van Loan method for Sylvester equations. This closed form makes implementation straightforward: the integral depends only on the horizon $T - t$, not on the current state.

4.4. Implementation remarks. The pricing formulas above are tractable for three reasons. First, the kernels in (??)–(??) are computed from matrix exponentials, which are numerically stable and fast to evaluate. Second, the transform (??) depends only on one-dimensional integrals of Lévy exponents, so Fourier pricing reduces to low-dimensional numerical integration. Third, the same state-space representation supports exact or semi-exact simulation on a time grid via matrix exponentials, which is useful for Monte Carlo validation and for hedging backtests.

For calibration, the model remains close to standard CARMA practice. Seasonal components Λ_S, Λ_Y can be estimated separately; the stationary state equations can then be fitted by likelihood, quasi-likelihood, moment matching, or state-space filtering, depending on the driving noise. The cross-covariance formula (??) is especially useful for identifying the coupling matrix Γ .

5. HEDGING IN AN INCOMPLETE MARKET

5.1. Variance-optimal hedging with one traded forward. Because the secondary factor is typically non-tradable, the quanto market is in general incomplete. The correct hedging question is therefore not whether the claim can be replicated exactly, but how much of its risk can be eliminated by trading liquid instruments.

Fix a maturity T and let $\tilde{F}_t = (B_t^d)^{-1} F(t, T)$ be the discounted forward used for dynamic hedging. Let

$$\tilde{H} = (B_T^d)^{-1} g(S_T, \hat{Y}_T)$$

be square-integrable under the pricing measure $\mathbb{Q}$ of ?????. When the secondary factor is non-tradable, one may alternatively work under the *minimal martingale measure* $\mathbb{Q}^{\min}$ (which preserves the law of non-tradable factors while making the traded forward a martingale) or under the *variance-optimal martingale measure*; see ?] for a systematic treatment. For concreteness, we develop the decomposition under $\mathbb{Q}$ and note that the formulas hold verbatim under any equivalent measure that makes $\tilde{F}$ and $\tilde{V}$ square-integrable martingales. Define the discounted value process

$$\tilde{V}_t = \mathbb{E}^{\mathbb{Q}}[\tilde{H} \mid \mathcal{F}_t].$$

Lemma 5.1 (Square-integrability of the claim and the forward). *Under ?? and ??, with the payoff class (??) where h is polynomially bounded:*

- (a) *The discounted forward $\tilde{F}_t = (B_t^d)^{-1} F(t, T)$ is a square-integrable $\mathbb{Q}$ -martingale.*
- (b) *The discounted claim $\tilde{H} = (B_T^d)^{-1} g(S_T, \hat{Y}_T)$ satisfies $\tilde{H} \in L^2(\mathbb{Q})$.*

Proof. Part (a). By the tower property $F(t, T)^2 \leq \mathbb{E}^{\mathbb{Q}}[S_T^2 \mid \mathcal{F}_t]$ (Jensen applied to the square, since $S_T = \mathbb{E}^{\mathbb{Q}}[S_T \mid \mathcal{F}_t]$). We compute $\mathbb{E}^{\mathbb{Q}}[S_T^2 \mid \mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}}[e^{2X_T} \mid \mathcal{F}_t]$ (absorbing the deterministic seasonal factor). Setting $u = -2i$ in (??) and applying the same analytic continuation as in ?? with $\alpha_1 = 2$ shows that

$$\mathbb{E}^{\mathbb{Q}}[e^{2X_T} \mid \mathcal{F}_t] = \exp\left(2\bar{X}_t(T) + \int_t^T \kappa_X^{\mathbb{Q}}(2\beta_X(s; T)) ds + \int_t^T \kappa_Y^{\mathbb{Q}}(2\gamma_X(s; T)) ds\right) < \infty$$

whenever $2\beta_X(s; T) \in \text{int}(\mathcal{S}_X)$ and $2\gamma_X(s; T) \in \text{int}(\mathcal{S}_Y)$ for all $s \in [t, T]$; these inclusions hold under (2.10) by the kernel bounds of (2.10) applied with the doubled amplitude. Hence $\tilde{F} \in L^2(\mathbb{Q})$. That $\tilde{F}$ is a $\mathbb{Q}$ -martingale follows from the tower property and the fact that $F(t, T) = \mathbb{E}^\mathbb{Q}[S_T | \mathcal{F}_t]$.

Part (b). For $g(S_T, \tilde{Y}_T) = (S_T - K_S)^+ h(\tilde{Y}_T)$ one has $(S_T - K_S)^+ \leq S_T$, so $\tilde{H}^2 \leq (B_T^d)^{-2} S_T^2 h(\tilde{Y}_T)^2$. Since h is polynomially bounded, $h(\tilde{Y}_T)^2 \leq C(1 + |Y_T|^k)^2$ for some constants $C, k > 0$. By Cauchy-Schwarz, $\mathbb{E}^\mathbb{Q}[\tilde{H}^2] \leq (B_T^d)^{-2} \sqrt{\mathbb{E}^\mathbb{Q}[S_T^4]} \sqrt{\mathbb{E}^\mathbb{Q}[(1 + |Y_T|^k)^4]}$. The first factor is finite by part (a) with $\alpha_1 = 4$ (using exponential moments on X_T from (2.10)); the second factor is finite because $\mathbb{E}^\mathbb{Q}[|Y_T|^{4k}]$ is finite whenever $\kappa_Y^\mathbb{Q}$ has finite derivatives of order $4k$ at the origin, which holds under the exponential moment condition in (2.10). $\square$

Proposition 5.2 (Galtchouk–Kunita–Watanabe decomposition). *Assume that $\tilde{F}$ is a square-integrable $\mathbb{Q}$ -martingale (2.10) and that $\tilde{H} \in L^2(\mathbb{Q})$ (2.10). Then there exist a predictable process ξ and a square-integrable $\mathbb{Q}$ -martingale $M^\perp$ with $M_0^\perp = 0$ and $\langle M^\perp, \tilde{F} \rangle \equiv 0$ such that*

$$\tilde{V}_t = \tilde{V}_0 + \int_0^t \xi_u d\tilde{F}_u + M_t^\perp, \quad 0 \leq t \leq T. \quad (21)$$

The predictable process ξ is the unique $L^2(\mathbb{Q})$ -optimal hedge: it minimizes

$$\mathbb{E}^\mathbb{Q} \left[\left(\tilde{H} - \tilde{V}_0 - \int_0^T \theta_u d\tilde{F}_u \right)^2 \right]$$

over all predictable θ , and it satisfies

$$\xi_t = \frac{d\langle \tilde{V}, \tilde{F} \rangle_t}{d\langle \tilde{F} \rangle_t} \quad \text{whenever } d\langle \tilde{F} \rangle_t > 0. \quad (22)$$

Proof. The GKW decomposition is the classical orthogonal projection in the Hilbert space of square-integrable martingales; see [1] and the references therein. Any square-integrable martingale can be uniquely written as an initial value, a stochastic integral with respect to the reference martingale $\tilde{F}$, and a martingale orthogonal to $\tilde{F}$. Existence and uniqueness follow from the L^2 projection theorem. Orthogonality of $M^\perp$ to $\tilde{F}$ means that the stochastic integral term captures the full covariation of $\tilde{V}$ with $\tilde{F}$, hence it minimizes the residual variance. Formula (22) is the predictable Radon–Nikodym representation of the projection coefficient, valid when the quadratic variation of $\tilde{F}$ is absolutely continuous in time. $\square$

Remark 5.3 (Admissibility of the hedge). *The strategy ξ produced by (22) is admissible in the sense that $\int_0^T \xi_u^2 d\langle \tilde{F} \rangle_u < \infty$ $\mathbb{Q}$ -a.s. This follows from $\tilde{V}, \tilde{F} \in L^2(\mathbb{Q})$ (2.10) together with the Cauchy-Schwarz inequality applied to the Radon-Nikodym representation (22): by the Kunita-Watanabe inequality, $\int_0^T \xi_u^2 d\langle \tilde{F} \rangle_u \leq \langle \tilde{V} \rangle_T \leq \mathbb{E}^\mathbb{Q}[\tilde{H}^2] < \infty$.*

Remark 5.4. *We use the notation $M^\perp$ for the orthogonal residual martingale to avoid confusion with the Lévy drivers L^X, L^Y used in the state-space dynamics.*

Remark 5.5. *Equation (22) provides the right interpretation of “delta hedging” in an energy quanto problem. The hedge ratio ξ_t is not merely the derivative of a pricing formula with respect to the spot; it is the predictable projection of the claim’s martingale part onto the traded forward. The residual martingale $M^\perp$ is not a modeling nuisance but the mathematically precise unhedgeable quanto risk. Its terminal value $M_T^\perp$ quantifies the irreducible tracking error under the chosen measure $\mathbb{Q}$.*

5.2. Multiple instruments and proxy hedges. In practice one often has more than one liquid hedging instrument: a strip of power forwards, a gas forward, an FX forward, or sometimes a weather future or a proxy contract. Let

$$\tilde{F}_t = (\tilde{F}_t^1, \dots, \tilde{F}_t^m)^\top$$

be the vector of discounted traded instruments. The same projection principle gives the multi-asset hedge.

Proposition 5.6 (Multi-asset variance-optimal hedge). *Assume that $\tilde{F}$ is an m -dimensional square-integrable $\mathbb{Q}$ -martingale and that the predictable covariance matrix*

$$\Sigma_t^{FF} = \left(\frac{d\langle \tilde{F}^i, \tilde{F}^j \rangle_t}{dt} \right)_{1 \leq i, j \leq m}$$

is invertible almost everywhere. Then the variance-optimal hedge vector $\xi_t \in \mathbb{R}^m$ is

$$\xi_t = (\Sigma_t^{FF})^{-1} \Sigma_t^{FV}, \quad \Sigma_t^{FV} := \left(\frac{d\langle \tilde{F}^i, \tilde{V} \rangle_t}{dt} \right)_{1 \leq i \leq m}. \quad (23)$$

Proof. This is the vector version of the Galtchouk–Kunita–Watanabe projection. $\square$

Remark 5.7. *The multivariate hedge (??) is precisely where an MCARMA formulation can be more than a matter of elegance. If the secondary factor is a tradable FX rate, or if a weather future or proxy instrument is available, then the covariance matrix in (??) can absorb a significant part of the residual hybrid risk. In the classical FX-quanto setting, if one of the instruments $\tilde{F}^i$ is an FX forward (tradable secondary factor), the residual $M^\perp$ is substantially reduced and the market becomes “almost complete” if the only remaining unspanned source is stochastic volatility or a spread risk. A fully multivariate state-space specification then provides the natural joint model for the tradable hedge universe. In contrast, when only the energy forward is liquid, the simpler coupled model is often preferable because it isolates the residual non-tradable exposure explicitly.*

5.3. Practical hedge decomposition. The CARMA framework suggests a useful decomposition of hedgeable and unhedgeable risk:

- (i) *Energy price risk:* the component projected onto traded forwards or futures. This is the part captured by $\int \xi d\tilde{F}$.
- (ii) *Cross-factor risk:* the part created by the coupling matrix Γ . It affects the hedge ratio through the predictable covariation of the claim value with the traded instrument.
- (iii) *Pure secondary-factor risk:* the orthogonal component in $M^\perp$, which remains after projection and is unavoidable unless additional tradable instruments are introduced.

In applications this decomposition is more informative than a list of Greeks. It tells the practitioner not only how to hedge, but also which risk cannot be hedged within the available market.

5.4. Explicit hedge ratio for the Brownian case. The abstract formula (??) becomes explicit in the coupled CARMA framework. We state the result for the Brownian specialization of ?? (continuous-path drivers), in which the quadratic covariations are deterministic and computable from matrix exponentials.

Proposition 5.8 (CARMA hedge ratio). *Suppose that both $L^{X,\mathbb{Q}}$ and $L^{Y,\mathbb{Q}}$ are Brownian motions (possibly multivariate). Let $\Sigma_X = \text{Var}^{\mathbb{Q}}(L_1^{X,\mathbb{Q}})$ and $\Sigma_Y = \text{Var}^{\mathbb{Q}}(L_1^{Y,\mathbb{Q}})$. Then the forward price satisfies*

$$d\tilde{F}_t = \tilde{F}_t \left(\beta_X(t; T)^\top dW_t^X + \gamma_X(t; T)^\top dW_t^Y \right), \quad (24)$$

where W^X and W^Y are the independent Brownian drivers of $L^{X,\mathbb{Q}}$ and $L^{Y,\mathbb{Q}}$ respectively, and the instantaneous quadratic variation of $\tilde{F}$ is

$$\frac{d\langle \tilde{F} \rangle_t}{dt} = \tilde{F}_t^2 \left(\beta_X(t; T)^\top \Sigma_X \beta_X(t; T) + \gamma_X(t; T)^\top \Sigma_Y \gamma_X(t; T) \right) =: \tilde{F}_t^2 \sigma_F^2(t; T). \quad (25)$$

By the Markov property of the coupled state (Z_t^X, Z_t^Y) , the discounted option value satisfies $\tilde{V}_t = V(t, Z_t^X, Z_t^Y)$ for a smooth pricing function V . Applying Itô's formula and using the state equations (??):

$$d\tilde{V}_t = (\nabla_{Z^X} V_t)^\top B_X dW_t^X + (\nabla_{Z^X} V_t)^\top \Gamma dW_t^Y + (\nabla_{Z^Y} V_t)^\top B_Y dW_t^Y + (\text{drift}) dt, \quad (26)$$

where $\nabla_{Z^X} V_t := \nabla_{Z^X} V(t, Z_t^X, Z_t^Y) \in \mathbb{R}^{p_X}$ and $\nabla_{Z^Y} V_t := \nabla_{Z^Y} V(t, Z_t^X, Z_t^Y) \in \mathbb{R}^{p_Y}$ are the state-space gradients of the pricing function. The predictable cross-variation with $\tilde{F}$ is therefore

$$\frac{d\langle \tilde{V}, \tilde{F} \rangle_t}{dt} = \tilde{F}_t \left[(\nabla_{Z^X} V_t)^\top B_X \Sigma_X \beta_X(t; T) + (\nabla_{Z^X} V_t)^\top \Gamma \Sigma_Y \gamma_X(t; T) + (\nabla_{Z^Y} V_t)^\top B_Y \Sigma_Y \gamma_X(t; T) \right]. \quad (27)$$

Substituting into (??) and cancelling $\tilde{F}_t$ gives the explicit variance-optimal hedge ratio:

$$\xi_t = \frac{(\nabla_{Z^X} V_t)^\top B_X \Sigma_X \beta_X(t; T) + (\nabla_{Z^X} V_t)^\top \Gamma \Sigma_Y \gamma_X(t; T) + (\nabla_{Z^Y} V_t)^\top B_Y \Sigma_Y \gamma_X(t; T)}{\beta_X(t; T)^\top \Sigma_X \beta_X(t; T) + \gamma_X(t; T)^\top \Sigma_Y \gamma_X(t; T)}. \quad (28)$$

The state gradients $\nabla_{Z^X} V_t$ and $\nabla_{Z^Y} V_t$ are computed by differentiating the Fourier pricing formula (??) with respect to the current conditional mean terms $\bar{X}_t(T)$ and $\bar{Y}_t(T)$, which appear linearly in the exponent of φ_t . Since $\bar{X}_t(T) = c_X^\top e^{A_X^{\mathbb{Q}}(T-t)} Z_t^X$ and $\bar{Y}_t(T) = c_Y^\top e^{A_Y^{\mathbb{Q}}(T-t)} Z_t^Y$, the chain rule gives

$$\nabla_{Z^X} V_t = \frac{B_t^d}{B_T^d} \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \widehat{g_{T,\alpha}}(u, v) i(u - i\alpha_1) e^{A_X^{\mathbb{Q}}(T-t)} c_X \varphi_t(u - i\alpha_1, v - i\alpha_2; T) du dv, \quad (29)$$

and analogously for $\nabla_{Z^Y} V_t$ with $(u, v) \mapsto i(v - i\alpha_2)$ and c_Y , $A_Y^{\mathbb{Q}}$. Differentiation under the integral is justified by dominated convergence under the same moment conditions as ??.

Proof. The forward SDE (??) follows from Itô's formula applied to $F(t, T) = \exp(\Lambda_S(T) + \bar{X}_t(T) + \int_t^T \kappa_X^{\mathbb{Q}}(\beta_X(s; T)) ds + \dots)$ from (??). Since the kernels β_X and γ_X are the sensitivities of the log-forward price to the increments of L^X and L^Y over $[t, T]$, the variation-of-constants formula (??) shows that the W^X and W^Y loadings of $d\bar{X}_t(T)$ are exactly $-c_X^\top e^{A_X^{\mathbb{Q}}(T-t)} B_X dW_t^X = -\beta_X(t; T)^\top \Sigma_X^{1/2} dW_t^X$ and $-\gamma_X(t; T)^\top \Sigma_Y^{1/2} dW_t^Y$ respectively; the drift term in Itô's formula for the exponential then produces the quadratic variation term, yielding (??). The quadratic variation (??) follows immediately from Itô's isometry. The option SDE (??) follows from Itô's formula applied to the smooth function $V(t, \cdot, \cdot)$, using the state equations (??) in the Brownian case. The cross-variation (??) is then the bilinear

covariation of (??) and (??), summing the contributions from W^X and W^Y . Formula (??) follows from differentiating (??) under the integral sign with respect to $\bar{X}_t(T)$; dominated convergence applies under the moment conditions of ??.

Remark 5.9. *In a Monte Carlo implementation, the hedge ratio can alternatively be estimated pathwise: simulate the coupled CARMA state over a time grid using matrix exponentials, compute the forward $F(t_k, T)$ and the option value estimate at each rebalancing date, and estimate the ratio (??) by regression of the value increments on the forward increments. This pathwise approach avoids the need to differentiate the Fourier formula and is robust to the choice of Lévy driver.*

Remark 5.10 (Extension to the general Lévy case). ?? is stated for Brownian drivers in order to obtain the explicit closed-form (??). When $L^{X, \mathbb{Q}}$ and $L^{Y, \mathbb{Q}}$ have non-trivial Lévy measures, the quadratic variation $\langle \tilde{F} \rangle_t$ and the cross-variation $\langle \tilde{V}, \tilde{F} \rangle_t$ each receive a jump contribution in addition to the Brownian term. Specifically, the predictable covariation density becomes

$$\begin{aligned} \frac{d\langle \tilde{V}, \tilde{F} \rangle_t^{\text{pred}}}{dt} = & \tilde{F}_t \left[(\nabla_{Z^X} V_t)^\top B_X \Sigma_X^c \beta_X(t; T) + (\nabla_{Z^X} V_t)^\top \Gamma \Sigma_Y^c \gamma_X(t; T) + (\nabla_{Z^Y} V_t)^\top B_Y \Sigma_Y^c \gamma_X(t; T) \right] \\ & + \int_{\mathbb{R}^{m_X}} \delta_X^V(t, x) \delta_X^F(t, x) \nu_X(dx) + \int_{\mathbb{R}^{m_Y}} \delta_Y^V(t, y) \delta_Y^F(t, y) \nu_Y(dy), \end{aligned} \quad (30)$$

where Σ_X^c and Σ_Y^c are the Gaussian covariance matrices of L^X and L^Y , and $\delta_X^V(t, x)$ (resp. $\delta_X^F(t, x)$) denotes the jump sensitivity of the option value (resp. the forward) to a jump of size x in L^X :

$$\delta_X^F(t, x) = \tilde{F}_t (e^{\beta_X(t; T)^\top x} - 1), \quad \delta_X^V(t, x) = V(t, Z_t^X + B_X x, Z_t^Y) - V(t, Z_t^X, Z_t^Y),$$

and analogously for jumps in L^Y . The variance-optimal hedge ratio retains the form (??) with $d(\cdot)_t$ replaced by the predictable covariation density (??). In the NIG case the Lévy jump sensitivities δ^V and δ^F can be evaluated efficiently via the Fourier formula (??) and its state-shifted counterparts; we refer to the numerical section for the implementation.

6. CALIBRATION TO GERMAN ENERGY MARKETS

The theoretical framework of Sections ??–?? is self-contained, but its practical value depends on whether the coupled CARMA structure fits real energy data better than simpler alternatives and whether the model parameters can be identified reliably from hourly observations. This section calibrates the model to hourly German day-ahead electricity prices and hourly temperature observations, implementing the calibration programme outlined in ??. All estimation code is available in the accompanying Python notebooks.

6.1. Data and preprocessing.

Price series. We use cleaned hourly German day-ahead prices from EPEXSPOT (DE/LU zone), covering the period January 2023 – December 2025. The raw file switches to quarter-hour observations late in 2025; these are aggregated to hourly averages before deseasonalization. The training window is January 2023 – December 2024 (17,544 hourly observations) and the 2025 holdout window contains 8,737 hourly observations. A distinctive feature of European electricity markets is the occurrence of negative prices during periods of high renewable feed-in and low demand. In the cleaned hourly sample the minimum observed price is -500 EUR/MWh. To work with a log-price factor we apply the constant shift

$$\tilde{S}_t = \log(S_t + \delta_{\text{shift}}), \quad \delta_{\text{shift}} = 1000 \text{ EUR/MWh}, \quad (31)$$

FIGURE 1. Deseasonalized temperature (top row) and log-price (bottom row). Left: raw series with estimated seasonal component $\Lambda(t)$ overlaid. Right: residual series Y_t (temperature) and X_t (log-price) after deseasonalization, together with their sample ACF at lags 0–96 h.

so that $\tilde{S}_t$ is always well-defined. In the empirical implementation the positive state variable is $S_t + \delta_{\text{shift}}$; reported market prices are recovered by subtracting the shift after valuation. The choice $\delta_{\text{shift}} = 1000$ ensures $S_t + \delta_{\text{shift}} > 0$ throughout the sample, giving a log-price series $\tilde{S}_t$ with sample mean 6.99 and standard deviation 0.047.

Temperature series. Hourly air temperature is obtained from Deutscher Wetterdienst (DWD) at a representative station in central Germany. The temperature training window is January 2020 – December 2024 (43,848 hourly observations) and the 2025 holdout window contains 8,736 hourly observations. The longer temperature history allows for a more precise seasonal fit and a more stable CARMA parameter estimate.

6.2. Deseasonalization. The observed log-price $\tilde{S}_t$ and temperature $\hat{Y}_t$ each contain strong deterministic components at the hourly, daily, and annual frequencies that must be removed before stochastic modeling. We fit a Fourier regression model of the form

$$\Lambda(t) = \alpha_0 + \sum_{k=1}^{K_{\text{hr}}} \left[a_k^{\text{hr}} \cos\left(\frac{2\pi kt}{24}\right) + b_k^{\text{hr}} \sin\left(\frac{2\pi kt}{24}\right) \right] + \sum_{k=1}^{K_{\text{yr}}} \left[a_k^{\text{yr}} \cos\left(\frac{2\pi kt}{8760}\right) + b_k^{\text{yr}} \sin\left(\frac{2\pi kt}{8760}\right) \right] + \text{interaction terms}, \quad (32)$$

where t is measured in hours, $K_{\text{hr}} = 3$ harmonic pairs capture the intraday shape, and $K_{\text{yr}} = 3$ pairs capture the annual cycle. The interaction terms are cross-products of intraday and annual harmonics, included to model the fact that the intraday shape differs between summer and winter. For the price series, we additionally include six day-of-week dummies.

The seasonal model is estimated by ordinary least squares on the training window only, then applied to the full sample. Stationarity of the residuals $X_t = \tilde{S}_t - \Lambda_S(t)$ and $Y_t = \hat{Y}_t - \Lambda_Y(t)$ is verified by augmented Dickey–Fuller tests, which reject the unit-root null at the 1% level for both series. ?? shows the raw and deseasonalized series for both factors. The autocorrelation functions of the residuals display clear but decaying serial dependence, motivating continuous-time autoregressive dynamics.

6.3. Marginal CARMA calibration. We fit competing CARMA(p, q) models to each residual series using maximum likelihood via a Kalman filter. The state-space model is discretised exactly using the Van Loan zero-order-hold formula: the transition matrix $F = e^{A\Delta t}$ and the discrete noise covariance Q_d are computed from the matrix exponential of a $2p \times 2p$ block matrix, guaranteeing numerical stability. The initial covariance is the stationary solution of the Lyapunov equation. The parameter vector $\theta = (a_1, \dots, a_p, b_0, \dots, b_{q-1}, \log \sigma)$ is optimised by L-BFGS-B with five random starting points derived from an ARMA moment-matching initialisation, and the best solution is retained. All eigenvalues of the fitted companion matrix are verified to have strictly negative real parts.

?? reports the log-likelihood, AIC, and BIC for a range of CARMA orders, together with an OU (CARMA(1, 0)) benchmark. For both series, the CARMA models with $p \geq 2$ improve substantially over the OU benchmark, confirming the need

TABLE 1. CARMA model comparison. Log-likelihood and information criteria for competing CARMA(p, q) specifications. The selected model (lowest AIC among those passing a Ljung-Box whiteness test at lag 24) is highlighted in bold. n denotes the number of parameters.

Series	Model	p	q	$\log \mathcal{L}$	AIC	BIC
Temperature	OU / CARMA(1, 0)	1	0	-61,240	122,484	122,501
Temperature	CARMA(2, 0)	2	0	-58,812	117,632	117,657
Temperature	CARMA(2, 1)	2	1	-58,706	117,422	117,455
Temperature	CARMA(3, 1)	3	1	-58,621	117,256	117,298
Temperature	CARMA(3, 2)	3	2	-58,618	117,252	117,302
Log-price	OU / CARMA(1, 0)	1	0	-22,108	44,220	44,232
Log-price	CARMA(2, 0)	2	0	-21,843	43,694	43,712
Log-price	CARMA(2, 1)	2	1	-21,790	43,590	43,614
Log-price	CARMA(3, 1)	3	1	-21,731	43,478	43,508
Log-price	CARMA(3, 2)	3	2	-21,730	43,480	43,516

TABLE 2. MLE parameter estimates for the selected CARMA(3, 1) models. Companion-matrix eigenvalues λ_i are in yr^{-1} ; mean-reversion times $\tau_i = 1/|\lambda_i|$ are given in years and hours. Standard errors (not shown) are obtained from the observed Fisher information.

Parameter	Temperature (Y)		Log-price (X)	
	Estimate	Unit	Estimate	Unit
a_1	210.3	yr^{-1}	3,842	yr^{-1}
a_2	18,440	yr^{-2}	1,127,000	yr^{-2}
a_3	504,100	yr^{-3}	8.4×10^7	yr^{-3}
b_0	1.24	—	0.38	—
σ	2.06	$\text{yr}^{-1/2}$	18.7	$\text{yr}^{-1/2}$
λ_1 (Re)	-8.1	yr^{-1}	-415	yr^{-1}
τ_1	0.124 yr	1,083 h	0.0024 yr	21.1 h
λ_2 (Re)	-101	yr^{-1}	-1,713	yr^{-1}
τ_2	0.0099 yr	86.7 h	0.00058 yr	5.1 h
λ_3 (Re)	-101	yr^{-1}	-1,713	yr^{-1}
τ_3	0.0099 yr	86.7 h	0.00058 yr	5.1 h

for higher-order continuous-time dynamics to capture the empirical autocovariance structure of German energy data.

?? reports the maximum-likelihood parameter estimates for the selected CARMA(3, 1) specifications. The eigenvalues of the companion matrix translate directly into mean-reversion time scales: the faster temperature eigenvalue (λ_1^Y) corresponds to a half-life of a few days while the slower one (λ_2^Y, λ_3^Y or complex pair) captures the persistent weekly component. For log-prices, the fastest eigenvalue reflects rapid spike reversion (of the order of hours), consistent with the hourly resolution of the data and the well-documented mean-reverting jump structure of European power prices.

?? shows the standardised Kalman innovations for each fitted model together with their sample ACF and QQ-plots. The Ljung-Box p -values at lags 24, 48, and 72 exceed 0.05 for the selected models, indicating that the innovations are

FIGURE 2. Standardised Kalman innovations for the selected CARMA models. Left: observed versus one-step-ahead prediction (first 1,000 observations). Centre: ACF of standardised innovations (lags 0–72 h); dashed lines show the 95% confidence band. Right: QQ-plot against $\mathcal{N}(0, 1)$.

TABLE 3. Gaussian versus NIG distribution fit to recovered Lévy increments. $\Delta\text{AIC} = \text{AIC}_{\text{Gauss}} - \text{AIC}_{\text{NIG}}$ (positive favours NIG). Excess kurtosis of sample increments is reported in the last column.

Series	Model	$\log \mathcal{L}$	AIC	BIC	ΔAIC	Kurt.(excess)
Temperature	Gaussian	−31,820	63,644	63,657	—	3.8
Temperature	$\text{NIG}(\alpha, \beta, \mu, \delta)$	−26,104	52,216	52,244	11,428	—
Log-price	Gaussian	−22,508	45,020	45,031	—	42.1
Log-price	$\text{NIG}(\alpha, \beta, \mu, \delta)$	−20,916	41,840	41,862	3,180	—

approximately uncorrelated. Some residual non-Gaussianity is visible in the QQ-plots, particularly heavy tails for the log-price innovations, which motivates the Lévy noise specification explored in the next subsection.

6.4. Lévy noise characterization. The Lévy increments dL_t^Y and dL_t^X are recovered from the Kalman-filtered state estimates using the exact propagator formula

$$\Delta L_k = \frac{B_{\text{load}}^\top (Z_{k+1} - FZ_k)}{\|B_{\text{load}}\|^2}, \quad B_{\text{load}} = A^{-1}(F - I)B, \quad F = e^{A\Delta t}, \quad (33)$$

which inverts the exact discrete-time propagator without approximation. We then fit Normal Inverse Gaussian (NIG) and Gaussian distributions to the recovered increments by maximum likelihood and compare them using AIC and BIC.

The NIG distribution is parameterised as $\text{NIG}(\alpha, \beta, \mu, \delta)$, where $\alpha > 0$ controls tail heaviness, $|\beta| < \alpha$ controls skewness, μ is a location parameter, and $\delta > 0$ is a scale parameter; see [?] for background. Its characteristic exponent is

$$\psi_{\text{NIG}}(u) = i\mu u + \delta \left(\sqrt{\alpha^2 - \beta^2} - \sqrt{\alpha^2 - (\beta + iu)^2} \right),$$

which enters the joint characteristic function (??) directly.

?? reports the model comparison. The NIG specification achieves a substantially better log-likelihood than the Gaussian in both series. For the log-price increments, the AIC improvement exceeds 3,000 points, consistent with the well-documented heavy tails of electricity price increments. Temperature increments also display modest but statistically significant excess kurtosis. ?? illustrates the density fits in the tails.

?? reports the fitted NIG parameters. The estimated α parameters reflect the distinct tail behaviour of the two series: temperature increments are closer to Gaussian (larger relative α , small $|\beta|$), while log-price increments are severely leptokurtic with visible negative skewness ($\beta < 0$), consistent with the asymmetry between large positive spikes and moderate negative corrections in German electricity prices.

6.5. Coupling estimation. To estimate the coupling parameter Γ we align the temperature and log-price increment series on their common observation window (2023–2024), obtaining $n = 17,519$ contemporaneous hourly pairs $(\Delta L_k^Y, \Delta L_k^X)$. ?? shows the empirical cross-correlation function (CCF) at lags $-72, \dots, +72$ hours, plotted against the $\pm 1.96/\sqrt{n}$ confidence band.

TABLE 4. NIG maximum-likelihood estimates. The NIG moment formulas give mean = $\mu + \delta\beta/\gamma$, variance = $\delta\alpha^2/\gamma^3$, skewness = $3\beta/(\alpha\sqrt{\delta\gamma})$, and excess kurtosis = $3(1 + 4\beta^2/\alpha^2)/(\delta\gamma)$, where $\gamma = \sqrt{\alpha^2 - \beta^2}$.

Series	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\mu}$	$\hat{\delta}$	n_{obs}
Temperature	6.83	-0.12	-0.004	0.361	43,823
Log-price	1.24	-0.21	-0.008	0.072	17,519

FIGURE 3. Distribution fit to recovered log-price Lévy increments. Left: histogram with Gaussian (blue) and NIG (red) density overlaid. Centre: log-density on $[-6, 6]$; the heavy tails visible on a linear scale become clearly sub-Gaussian on the log scale for the NIG fit. Right: QQ-plot of standardised increments versus $\mathcal{N}(0, 1)$.

FIGURE 4. Empirical CCF between temperature increments ΔL^Y and log-price increments ΔL^X at lags -72 to $+72$ h. Dashed lines show the 95% asymptotic confidence band $\pm 1.96/\sqrt{n}$.

The contemporaneous cross-correlation is 0.11 (t -statistic 14.2, $p < 0.001$), confirming a statistically significant directional link from temperature to electricity price at zero lag. The CCF is negligible at most other lags, consistent with the directional coupled CARMA specification in (??) in which the coupling enters exclusively at the instantaneous level through the shared noise channel.

We estimate the scalar coupling strength by OLS:

$$\hat{\gamma}_0 = \frac{\widehat{\text{Cov}}(\Delta L^X, \Delta L^Y)}{\widehat{\text{Var}}(\Delta L^Y)} = 0.094, \quad \text{SE}(\hat{\gamma}_0) = 0.0066, \quad R^2 = 0.012.$$

The coupling is economically small in terms of explained variance ($R^2 \approx 1.2\%$) but highly significant statistically. In the state-space notation of ??, the coupling matrix is $\Gamma = \hat{\gamma}_0 B_X$, so the price CARMA is driven by the linear combination $dL^X + \hat{\gamma}_0 dL^Y$. The residual $\varepsilon_k = \Delta L_k^X - \hat{\gamma}_0 \Delta L_k^Y$ is uncorrelated with ΔL^Y to three significant figures, confirming that the OLS estimate successfully removes the shared component.

7. NUMERICAL EXPERIMENTS

With the model fully calibrated to German market data, we turn to three numerical experiments that validate the theoretical results of ????: the accuracy of the Fourier pricing formula against Monte Carlo simulation, the economic significance of the quanto adjustment, and the out-of-sample hedging performance of the CARMA model relative to an OU benchmark.

7.1. Pricing accuracy: Fourier versus Monte Carlo. We compare the Fourier valuation formula (??) against a Monte Carlo benchmark for three classes of quanto payoffs:

- (i) *Plain energy call*: $g(S_T, Y_T) = (S_T - K_S)^+$;
- (ii) *Indicator quanto*: $g(S_T, Y_T) = \mathbf{1}_{\{S_T > K_S\}} \mathbf{1}_{\{Y_T > K_Y\}}$;
- (iii) *Linear quanto call*: $g(S_T, Y_T) = (S_T - K_S)^+ \cdot (Y_T - K_Y)$;

for a range of strikes (K_S, K_Y) and maturities $T \in \{1 \text{ week}, 1 \text{ month}, 3 \text{ months}\}$.

TABLE 5. Fourier versus Monte Carlo pricing for three quanto pay-offs. Strikes are given as fraction of the current forward price (K_S/F) and the current secondary factor level (K_Y/Y_0). MC prices are based on $N = 100,000$ paths; 95% CI half-widths are in parentheses. The relative error is $|V_{\text{FFT}} - V_{\text{MC}}|/V_{\text{MC}}$.

Payoff	T	K_S/F	K_Y/Y_0	V_{FFT}	V_{MC} (95% CI)
Plain call	1 wk	0.95	—	0.0412	0.0410 (± 0.0003)
Plain call	1 mo	1.00	—	0.0638	0.0635 (± 0.0004)
Plain call	3 mo	1.05	—	0.0521	0.0518 (± 0.0004)
Indicator quanto	1 wk	1.00	1.00	0.2341	0.2335 (± 0.0008)
Indicator quanto	1 mo	1.00	1.00	0.2289	0.2282 (± 0.0008)
Indicator quanto	3 mo	1.00	1.00	0.2201	0.2195 (± 0.0009)
Linear quanto call	1 mo	1.00	—	0.0071	0.0069 (± 0.0002)
Linear quanto call	3 mo	1.00	—	0.0143	0.0140 (± 0.0003)

Monte Carlo implementation. The coupled CARMA state is simulated on a daily grid using the exact matrix-exponential propagator (Section 8 of `carma_utils.py`):

$$Z_{k+1}^X = F_X Z_k^X + B_X^{\text{load}} \Delta L_k^X + \Gamma^{\text{load}} \Delta L_k^Y, \quad F_X = e^{A_X \Delta t}, \quad B_X^{\text{load}} = A_X^{-1}(F_X - I)B_X, \quad (34)$$

and analogously for Z^Y . The increments $\Delta L^X, \Delta L^Y$ are drawn independently from NIG distributions with the fitted parameters of ??, using the normal variance-mean mixture $\Delta L = \mu + \beta V + \sqrt{V} Z$ with $V \sim \text{InvGauss}(\delta/\gamma, \delta^2)$ and $Z \sim \mathcal{N}(0, 1)$. We use $N = 100,000$ paths for the Monte Carlo estimates. Confidence intervals at level 95% are $\pm 1.96 \hat{\sigma}/\sqrt{N}$.

Fourier implementation. The joint characteristic function (??) is evaluated via Gauss-Legendre quadrature ($n = 64$ nodes over $[0, \tau]$) applied to the integrated Lévy exponents. For the plain call, the one-dimensional Carr-Madan FFT [?] is used with damping $\alpha = 1.5$. For the indicator quanto, the two-dimensional Fourier formula with factored payoff transform is evaluated by double quadrature.

?? reports the results for the CARMA(3, 1) model with NIG driving noise and coupling $\hat{\gamma}_0 = 0.094$. Fourier and Monte Carlo prices agree to within two standard deviations in all cases, confirming that the numerical implementation of the analytical formula is consistent with the simulation-based benchmark. The Fourier method is 50–200 times faster than the Monte Carlo at the same accuracy, making it practical for real-time hedging applications.

7.2. Economic magnitude of the quanto adjustment. The first-order quanto adjustment of ?? isolates the contribution of the coupling channel to option values. For the heating quanto payoff $g = (S_T - K_S)^+(K_Y - \hat{Y}_T)^+$, the adjustment (??) is

$$\left. \frac{\partial V^\lambda}{\partial \lambda} \right|_{\lambda=0} = - \underbrace{\frac{B_t^d}{B_T^d} \mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{S_T > K_S} | \mathcal{F}_t]}_{\text{digital-weighted call}} \cdot \underbrace{\int_t^T c_X^\top e^{A_X(T-u)} \Gamma B_Y^\top e^{A_Y^\top(T-u)} c_Y du}_{\text{kernel covariance}} \cdot \underbrace{\mathbb{Q}(\hat{Y}_T < K_Y | \mathcal{F}_t)}_{\text{secondary put probability}}.$$

We compute this adjustment as a function of maturity $\tau = T - t$ and the at-the-money strike $K_S = F(t, T)$, using the fitted system matrices and $\lambda = \hat{\gamma}_0$. ?? reports the full (Fourier) price, the independent-factor price ($\lambda = 0$), and the adjustment, together with its contribution as a percentage of the full price.

The quanto adjustment is negative and grows in absolute magnitude with maturity, consistent with (??): a positive temperature shock increases expected electricity prices, shifting the call into the money at precisely those states where the

TABLE 6. Quanto price decomposition. $V(\hat{\gamma}_0)$ is the Fourier price with estimated coupling. $V(0)$ is the price under independence. The adjustment $\Delta V = V(\hat{\gamma}_0) - V(0)$ is expressed both in price units (EUR/MWh) and as a percentage of $V(0)$. All results are for at-the-money heating quanto calls $g = (S_T - F)^+(K_Y - \hat{Y}_T)^+$ at current states.

Maturity τ	$V(\hat{\gamma}_0)$	$V(0)$	ΔV	$\Delta V/V(0)$
1 week	2.14	2.21	-0.07	-3.2%
1 month	4.83	5.02	-0.19	-3.8%
3 months	8.71	9.14	-0.43	-4.7%
6 months	11.2	11.9	-0.70	-5.9%

TABLE 7. Out-of-sample hedging performance on the 2025 test year. Weekly rebalancing; indicator quanto with 1-month maturity and at-the-money strikes. $VRR = 1 - \text{Var}(\Pi)/\text{Var}(\Delta V)$; RMSE is the root-mean-square weekly hedge error.

Strategy	VRR	RMSE (EUR/MWh)	Mean error
Unhedged	—		0.0182
OU hedge	28%		0.0155
CARMA(3, 1) hedge	41%		0.0135

heating leg $(K_Y - \hat{Y}_T)^+$ is out of the money. The magnitude (3–6%) is economically significant and confirms that ignoring the coupling channel leads to a systematic over-pricing of heating quanto contracts.

7.3. Hedging performance. We assess the variance-optimal hedge of ?? in an out-of-sample backtest on the 2025 test year. The hedged position is a short indicator quanto $H = \mathbf{1}_{\{S_T > K_S\}} \mathbf{1}_{\{Y_T > K_Y\}}$ with maturity $T = 1$ month, at-the-money strikes, and weekly rebalancing. The hedge instrument is a one-month energy forward (position ξ_t as in ??).

Three strategies are compared:

- (i) *Unhedged*: hold the short quanto without any forward hedge;
- (ii) *OU hedge*: compute the delta-hedge ratio from a calibrated OU (CARMA(1, 0)) model;
- (iii) *CARMA hedge*: use the GKW ratio ξ_t computed via finite differences on the Fourier price (?), with the calibrated CARMA(3, 1)-NIG model.

At each rebalancing date t_k the Kalman filter delivers state estimates $(Z_{t_k}^X, Z_{t_k}^Y)$. The forward price $F(t_k, T)$ is computed via (??) and the hedge ratio ξ_{t_k} via finite differences on the Fourier option price. The weekly hedge PnL from date t_k to t_{k+1} is

$$\Pi_{k+1} = V_{t_{k+1}} - V_{t_k} - \xi_{t_k}(F_{t_{k+1}} - F_{t_k}).$$

Hedging effectiveness is measured by the variance reduction ratio $VRR = 1 - \text{Var}(\Pi)/\text{Var}(V_{t_{k+1}} - V_{t_k})$.

The CARMA hedge achieves a variance reduction of 41% versus the unhedged position, compared to 28% for the OU benchmark, a gain of 13 percentage points. The improvement is consistent with the CARMA model's ability to capture both the multi-scale mean reversion of electricity prices (fast spike + slow persistence) and the coupling channel, which shifts the hedge ratio at the boundary of the in-the-money region. The residual unhedged variance (59%) is the irreducible quanto risk

FIGURE 5. Cumulative hedging PnL on the 2025 test year. Solid blue: unhedged short quanto position. Dashed orange: OU-hedged position. Solid red: CARMA(3,1)-NIG hedged position. The shaded band is the unhedged ± 1 std. dev. envelope.

$M^\perp$ in the GKW decomposition (??), arising from the non-tradable temperature factor and from the heavy-tailed Lévy increments.

8. LIMITATIONS AND EXTENSIONS

The framework developed above is analytically tractable, but it does not remove the standard difficulties of hybrid derivative modeling.

Pricing measure for the non-tradable factor. The choice of pricing measure for the non-tradable factor is not implied by no-arbitrage. Any serious empirical implementation must therefore defend the risk-premium specification for the secondary factor, either by calibration to hybrid OTC quotes or by an explicit actuarial or utility-based criterion. In practice, the Esscher transform of the secondary factor Lévy triplet offers a one-parameter family of equivalent measures that preserves the CARMA state-space structure and can be calibrated to at-the-money volatilities of proxy instruments.

Linearity of the state-space. Linear CARMA dynamics are well suited to mean reversion and short-memory dependence, but they do not capture every stylized feature of energy markets. Regime switching, stochastic volatility, structural breaks, and severe non-linearities in renewable output may require model extensions. Lévy-driven CARMA factors alleviate part of this issue by accommodating jumps and heavy tails, but they remain linear state-space systems. A natural extension is to allow the noise-loading matrices B_X, B_Y or the coupling matrix Γ to be state-dependent (e.g., driven by a Markov chain), recovering a regime-switching CARMA hybrid.

Non-tradability of the electricity spot. Spot electricity is not itself tradable. The hedging theory must therefore be interpreted through forwards, futures, or swaps, and delivery-period mismatch may create additional basis risk. The framework accommodates this by working with forward prices (??) and by treating the hedge as a projection problem, but it does not eliminate basis risk induced by differences between spot and contract delivery horizons.

Identification and calibration. A full MCARMA generalization is not automatically superior. Although it offers a more symmetric and often more elegant representation, it introduces additional parameters and can be difficult to identify from the available data. The coupled specification should therefore be viewed not as a short-cut, but as a structural restriction that is useful when it is economically justified. Concretely, identification of the coupling matrix Γ requires cross-covariance observations at multiple lags (??); misspecification of the CARMA orders (p_X, q_X, p_Y, q_Y) is the most important source of model risk in practice.

Extensions. There are several natural extensions. One can allow for stochastic interest rates, multiple traded commodities, explicit FX settlement, or a factor structure for several weather stations. One can also replace the single directed coupling channel by a low-rank multivariate CARMA loading, which would retain much of the tractability while bridging the gap between the coupled and unrestricted MCARMA models. Finally, combining the CARMA state-space structure with a hidden Markov filter would enable sequential updating of the state vector and real-time hedge rebalancing, turning the present theoretical framework into an operational trading system.

9. CONCLUSION

We have developed and empirically validated a coherent framework for pricing and hedging quanto options in CARMA-based models. The central idea is to combine a tradable energy factor and a secondary hybrid-risk factor in a coupled CARMA state-space system, driven by independent NIG Lévy noises linked through a single directional coupling channel.

The analysis leads to four main conclusions. First, coupled CARMA models are a natural intermediate layer between univariate CARMA models and full M-CARMA systems: they preserve interpretability while embedding cleanly into the multivariate theory. For German hourly electricity prices and temperature, a CARMA(3, 1) model with NIG driving noise fits substantially better than the OU benchmark, with eigenvalues that identify both a fast spike-reversion component (half-life of order hours) and a slower persistent component (half-life of order days). Second, the Fourier valuation formula (??) agrees with Monte Carlo to within simulation error across all payoff types and maturities, confirming that the analytical joint characteristic function (??) is correctly implemented and computationally efficient. Third, the coupling parameter $\hat{\gamma}_0 \approx 0.094$ is statistically highly significant and generates a quanto adjustment of 3–6% for heating-type payoffs, confirming that ignoring temperature–electricity dependence leads to a systematic pricing bias. Fourth, the variance-optimal GKW hedge reduces tracking-error variance by 41% with the CARMA model versus 28% with the OU benchmark, demonstrating the practical value of higher-order continuous-time dynamics and the explicit coupling channel for hedging hybrid claims.

The primary avenue for future work is the calibration of the pricing measure for the non-tradable factor. The present analysis uses the historical law of the secondary factor as a proxy; incorporating an Esscher or utility-based risk premium for temperature risk, calibrated to hybrid OTC quotes, would complete the bridge to a full mark-to-model implementation. Extending the hedging universe to weather futures or proxy instruments would further reduce the residual hybrid risk $M^\perp$.

REFERENCES

- [] Aurélien Alfonsi and Nerea Vadillo. Risk valuation of quanto derivatives on temperature and electricity, 2024. URL <https://arxiv.org/abs/2310.07692>.
- [] Fred Espen Benth and Sven Karbach. Multivariate continuous-time autoregressive moving-average processes on cones. *Stochastic Processes Appl.*, 162: 299–337, 2023. ISSN 0304-4149. doi: 10.1016/j.spa.2023.05.003.
- [] Fred Espen Benth and Che Mohd Imran Che Taib. On the speed towards the mean for continuous time autoregressive moving average processes with applications to energy markets. *Energy Economics*, 40:259–268, 2013. ISSN 0140-9883. doi: <https://doi.org/10.1016/j.eneco.2013.07.007>. URL <https://www.sciencedirect.com/science/article/pii/S0140988313001527>.
- [] Fred Espen Benth, Claudia Klüppelberg, Gernot Müller, and Linda Vos. Futures pricing in electricity markets based on stable carma spot models. *Energy Economics*, 44:392–406, 2014. ISSN 0140-9883. doi: <https://doi.org/10.1016/j.eneco.2014.03.020>. URL <https://www.sciencedirect.com/science/article/pii/S014098831400067X>.
- [] Fred Espen Benth, Nina Lange, and Tor Åge Myklebust. Pricing and hedging quanto options in energy markets. *Journal of Energy Markets*, 8(1):1–35, March 2015. ISSN 1756-3607.
- [] P. J. Brockwell. Lévy-driven CARMA processes. *Ann. Inst. Stat. Math.*, 53(1):113–124, 2001. ISSN 0020-3157. doi: 10.1023/A:1017972605872.

- Peter Carr and Dilip B. Madan. Option valuation using the fast Fourier transform. *The Journal of Computational Finance*, 2(4):61–73, 1999. doi: 10.21314/JCF.1999.043.
- Rama Cont and Peter Tankov. *Financial Modelling with Jump Processes*. Chapman & Hall/CRC Financial Mathematics Series. Chapman & Hall/CRC, 2004. ISBN 1-5848-8413-4.
- John Hull. *Options, futures and other derivatives*. Pearson, 11 edition, 2021.
- Tina Marquardt and Robert Stelzer. Multivariate CARMA processes. *Stochastic Processes and their Applications*, 117(1):96–120, 2007. doi: 10.1016/j.spa.2006.05.014.
- Martin Schweizer. A guided tour through quadratic hedging approaches. In Elyes Jouini, Jakša Cvitanić, and Marek Musiela, editors, *Option Pricing, Interest Rates and Risk Management*, Handbooks in Mathematical Finance, pages 538–574. Cambridge University Press, 2001. doi: 10.1017/CBO9780511569708.016.

INFORMATICS INSTITUTE AND KORTEWEG-DE VRIES INSTITUTE FOR MATHEMATICS, UNIVERSITY OF AMSTERDAM, SCIENCE PARK 105–107, 1098 XG AMSTERDAM, NETHERLANDS
 Email address: vincegachon@gmail.com

INFORMATICS INSTITUTE AND KORTEWEG-DE VRIES INSTITUTE FOR MATHEMATICS, UNIVERSITY OF AMSTERDAM, SCIENCE PARK 105–107, 1098 XG AMSTERDAM, NETHERLANDS
 Email address: sven@karbach.org